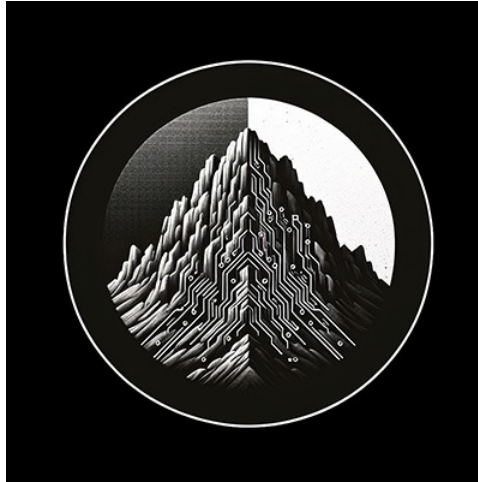




RTL Design Sherpa

Converters Micro-Architecture Specification 1.1

August 11, 2026

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2 Document Information

2.1 Document Control

Table 0.1: Document Control Information

Field	Value
Document Title	Converters Micro-Architecture Specification
Document Version	1.0
Component	Converters
Status	Active
Classification	Internal Technical
Last Updated	2026-01-03

2.2 Purpose

This Micro-Architecture Specification (MAS) describes how the Converters component is implemented — the internal detail behind the feature set. It covers:

- Internal block architectures
- State machine designs
- Signal timing and handshaking
- Resource utilization estimates
- Debug and verification strategies

2.3 Audience

This document is for:

- RTL designers implementing or modifying converter modules
- Verification engineers creating testbenches
- Integration engineers connecting converters to systems
- Performance engineers optimizing throughput and latency

2.4 Related Documents

Table 0.2: Related Documents

Document	Purpose
Converters Spec	High-level feature specification
Component PRD	Product requirements and goals
Bridge MAS	Related crossbar micro-architecture
Stream MAS	Related datapath micro-architecture

2.5 Revision History

Table 0.3: Revision History

Version	Date	Author	Description
1.0	2026-01-03	RTL Design Sherpa	Initial MAS release

2.6 Conventions

2.6.1 Notation

- `signal_name` - RTL signals and parameters
- **ModuleName** - Module names and key concepts
- *Figure X.X* - Figure references

2.6.2 Diagrams

All diagrams use Mermaid format rendered to PNG: - Source: `assets/mermaid/*.mmd` -
Rendered: `assets/mermaid/*.png`

2.6.3 Code Examples

The SystemVerilog snippets are implementation guidance: they show the intended design pattern but may differ slightly from the actual RTL.

Next: [Chapter 1: Introduction](#)

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3 Chapter 1: Introduction - Overview

3.1 1.1 Purpose

The Converters component provides data width and protocol conversion — the modules that let blocks with different data widths or communication protocols talk to each other.

3.2 1.2 Problem Statement

SoC designs run into the same two integration problems over and over:

3.2.1 1.2.1 Data Width Mismatch

Table 1.1: Common Data Width Configurations

Component	Typical Data Width
CPU	64-bit
DDR Controller	512-bit
PCIe Endpoint	128-bit
GPU	256-bit

Challenge: Mismatched widths can't be wired together directly; a width converter has to sit between them.

3.2.2 1.2.2 Protocol Incompatibility

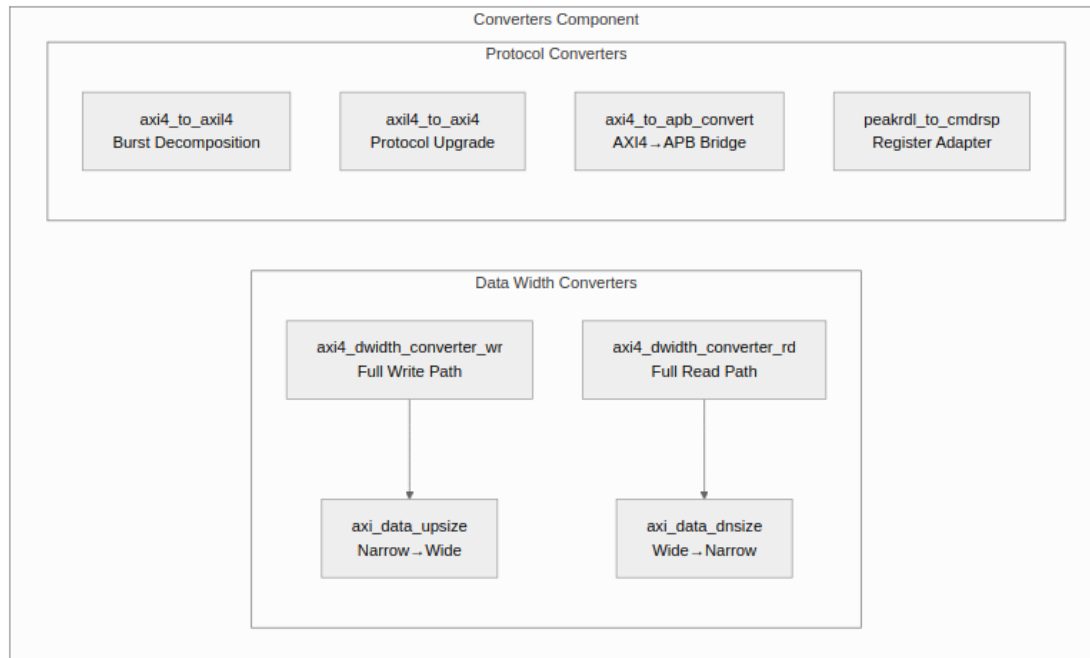
Table 1.2: Protocol Mismatch Examples

Master Type	Protocol	Slave Type	Protocol
CPU	AXI4	DDR	AXI4
DMA	AXI4	UART	APB
CPU	AXI4	GPIO	APB
Custom IP	AXI4-Lite	Fabric	AXI4

Challenge: Different protocols can't communicate without a bridge between them.

3.3 1.3 Solution Architecture

3.3.1 Figure 1.1: Converter Module Hierarchy



Converter Module Hierarchy

3.3.2 1.3.1 Data Width Converters

Generic Building Blocks: - **axi_data_upsize** - Accumulates N narrow beats into 1 wide beat - **axi_data_dsize** - Splits 1 wide beat into N narrow beats

Full AXI4 Converters: - **axi4_dwidth_converter_wr** - Complete write path (AW + W + B channels) - **axi4_dwidth_converter_rd** - Complete read path (AR + R channels)

3.3.3 1.3.2 Protocol Converters

AXI4 to AXI4-Lite: - **axi4_to_axil4_rd** - Read path burst decomposition - **axi4_to_axil4_wr** - Write path burst decomposition - **axi4_to_axil4** - Full bidirectional wrapper

AXI4-Lite to AXI4: - **axil4_to_axi4_rd** - Read path protocol upgrade - **axil4_to_axi4_wr** - Write path protocol upgrade - **axil4_to_axi4** - Full bidirectional wrapper

Other Converters: - **axi4_to_apb4_convert** - Full AXI4-to-APB bridge - **peakrdl_to_cmdrsp** - Register interface adapter

3.4 1.4 Key Design Decisions

3.4.1 1.4.1 Generic vs. Full Modules

The converters are built in three layers:

Layer 1: Generic Building Blocks

- `axi_data_upsize`, `axi_data_dnsize`
- Protocol-agnostic data manipulation
- Reusable across different contexts

Layer 2: Full AXI4 Converters

- `axi4_dwidth_converter_wr`, `axi4_dwidth_converter_rd`
- Compose generic blocks with AXI4 channel management
- Handle burst length adjustment, ID tracking

Layer 3: Protocol Converters

- `axi4_to_axil4`, `axi4_to_apb4`, etc.
- Full protocol translation
- State machine control

3.4.2 1.4.2 Throughput vs. Area Trade-offs

Table 1.3: Throughput vs. Area Trade-offs

Configuration	Throughput	Area	Use Case
Upsize (single buffer)	100%	1x	All narrow-to-wide
Downsize (single buffer)	80%	1x	Area-constrained
Downsize (dual buffer)	100%	2x	High-performance

Design Decision: Single-buffer upsize is always optimal (100% throughput at minimal cost). For downsize, the throughput/area trade-off is exposed as a configuration choice.

3.4.3 1.4.3 Sideband Signal Handling

Sideband signals (WSTRB, RRESP, etc.) support three handling modes:

Table 1.4: Sideband Handling Modes

Mode	Upsize Behavior	Downsize Behavior
Concatenate	Pack narrow strobes	Slice wide strobes
Broadcast	N/A	Repeat value

Mode	Upsize Behavior	Downsize Behavior
OR	Combine with OR	N/A

3.5 1.5 Performance Characteristics

3.5.1 1.5.1 Latency

Table 1.5: Latency Characteristics

Module	Single-Beat	Burst (N beats)
axi_data_upsize	0 cycles	N cycles to accumulate
axi_data_dnsize (single)	1 cycle	N cycles + gap
axi_data_dnsize (dual)	1 cycle	N cycles
axi4_to_axil4	0 cycles	2xN cycles
axil4_to_axi4	0 cycles	N/A (single only)

3.5.2 1.5.2 Throughput

Table 1.6: Throughput Characteristics

Module	Configuration	Peak Throughput
axi_data_upsize	Single buffer	1 beat/cycle
axi_data_dnsize	Single buffer	0.8 beats/cycle
axi_data_dnsize	Dual buffer	1 beat/cycle
axi4_to_axil4	Burst	0.5 beats/cycle
axil4_to_axi4	Any	1 beat/cycle

3.6 1.6 Scope

3.6.1 In Scope

- Integer width ratios (2:1, 4:1, 8:1, 16:1, etc.)
- AXI4, AXI4-Lite, and APB protocol support
- Configurable throughput vs. area trade-offs
- Generic building blocks for custom converters
- Burst-aware width conversion

3.6.2 Out of Scope

- Non-integer width ratios (e.g., 3:2 conversion)
- AXI4-Stream protocol (see Stream component)
- Complex buffering beyond dual-buffer
- Clock domain crossing (use separate CDC modules)
- Address translation (handled by crossbar)

3.7 1.7 Target Applications

1. **CPU-to-DDR Integration** - 64-bit CPU to 512-bit memory controller
 2. **DMA Engines** - Variable width data movers
 3. **Mixed Protocol Systems** - AXI4 fabric with APB peripheral bus
 4. **FPGA Fabric Interfaces** - Width matching for IP integration
 5. **Register Access** - PeakRDL to custom control protocols
-

Next: [Chapter 2: Data Width Converter Blocks](#)

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4 2.1 Generic Building Blocks

Two generic building blocks handle the data movement: **axi_data_upsize** (narrow-to-wide) and **axi_data_dnsz** (wide-to-narrow). Both are protocol-agnostic — compose them into full AXI4 converters or use them directly in custom designs.

4.1 2.1.1 Module Hierarchy

Generic Building Blocks

```
|
+-- axi_data_upsize.sv      # Narrow-to-Wide accumulator
|   +-- Accumulator buffer
|   +-- Beat counter
|   +-- Sideband packing
|
+-- axi_data_dnsize.sv      # Wide-to-Narrow splitter
|   +-- Data buffer (single or dual)
|   +-- Beat counter
|   +-- Sideband extraction
|   +-- Optional burst tracking
```

4.2 2.1.2 Design Philosophy

4.2.1 Separation of Concerns

The generic blocks handle **only data manipulation**: - Data packing/unpacking - Sideband signal handling - Valid/ready handshaking

They do **not** handle: - Address manipulation - Burst length adjustment - ID tracking - Protocol-specific signals (ARLEN, AWLEN, etc.)

That separation buys: - Reuse in multiple contexts - Simpler verification - Cleaner composition into full converters

4.2.2 Interface Pattern

Both modules use a consistent valid/ready interface:

```
// Input (narrow for upsize, wide for dnsz)
input  logic          i_valid,
output logic          o_ready,
input  logic [DATA_WIDTH-1:0] i_data,
input  logic [SB_WIDTH-1:0] i_sideband,
input  logic          i_last,          // Optional

// Output (wide for upsize, narrow for dnsz)
output logic          o_valid,
input  logic          i_ready,
output logic [DATA_WIDTH-1:0] o_data,
output logic [SB_WIDTH-1:0] o_sideband,
output logic          o_last          // Optional
```

4.3 2.1.3 Width Ratio Calculation

Both modules calculate the width ratio at elaboration time:

```
localparam int RATIO = WIDE_WIDTH / NARROW_WIDTH;  
localparam int RATIO_LOG2 = $clog2(RATIO);
```

```
// Example: 64-bit to 512-bit  
// RATIO = 512 / 64 = 8  
// RATIO_LOG2 = 3
```

Constraints: - WIDE_WIDTH must be an integer multiple of NARROW_WIDTH - Minimum ratio: 2 -
Maximum ratio: Typically 16 (limited by timing)

4.4 2.1.4 Sideband Modes

4.4.1 Upsize Sideband Modes

Table 2.1: Upsize Sideband Modes

Mode	Parameter	Behavior	Use Case
Concatenate	SB_OR_MODE=0	Pack N narrow sidebands	WSTRB packing
OR	SB_OR_MODE=1	OR all narrow sidebands	Error aggregation

Concatenate Example (WSTRB):

8 beats of 8-bit WSTRB → 1 beat of 64-bit WSTRB

[0]: 0xFF → output[7:0] = 0xFF

[1]: 0xF0 → output[15:8] = 0xF0

...

[7]: 0x0F → output[63:56] = 0x0F

OR Example (Error flags):

8 beats of error flags → 1 beat with any error

[0]: 0 → output = 0

[1]: 1 → output = 1 (error detected)

...

[7]: 0 → output remains 1

4.4.2 Downsize Sideband Modes

Table 2.2: Downsize Sideband Modes

Mode	Parameter	Behavior	Use Case
Slice	SB_BROADCAST=0	Extract slice per beat	WSTRB extraction
Broadcast	SB_BROADCAST=1	Repeat full value	RRESP broadcast

Slice Example (WSTRB):

1 beat of 64-bit WSTRB → 8 beats of 8-bit WSTRB

input = 0xFF_F0..._0F

[0]: output = 0xFF (bits [7:0])

[1]: output = 0xF0 (bits [15:8])

...

[7]: output = 0x0F (bits [63:56])

Broadcast Example (RRESP):

1 beat of 2-bit RRESP → 8 beats of 2-bit RRESP
input = OKAY (2'b00)
[0-7]: output = OKAY (all beats get same response)

4.5 2.1.5 Performance Comparison

4.5.1 Figure 2.1: Performance Comparison

Performance Comparison

Performance Comparison

Table 2.3: Performance Comparison

Module	Mode	Throughput	Latency	Area
axi_data_u psize	Single	100%	N cycles	1x
axi_data_d nsize	Single	80%	1 cycle	1x
axi_data_d nsize	Dual	100%	1 cycle	2x

Why 80% for single-buffer downsize?

A single buffer can't load the next wide beat while it's still draining the current one. The pattern is: load, output N narrow beats, one gap cycle, load again. That gap costs $1/(N+1)$ of throughput — efficiency approaches 100% for large N but never reaches it.

Why 100% for dual-buffer downsize?

With two buffers, loading and draining overlap: while one buffer outputs, the other loads. There are no gap cycles.

4.6 2.1.6 Resource Utilization

4.6.1 Upsize Resources (NARROW=64, WIDE=512)

Accumulator buffer: 512 bits (output register)
Beat counter: 3 bits ($\text{clog}_2(8)$)
Sideband accumulator: 64 bits (WSTRB)
Control logic: ~50 LEs

Total: ~70-100 LEs, ~580 registers

4.6.2 Downsize Resources (WIDE=512, NARROW=64)

Single Buffer:

Data buffer: 512 bits
Beat counter: 3 bits
Sideband logic: ~20 LEs
Control logic: ~50 LEs

Total: ~70-100 LEs, ~520 registers

Dual Buffer:

Data buffers (2x): 1024 bits
Beat counters (2x): 6 bits
Sideband logic: ~40 LEs
Control logic: ~100 LEs (ping-pong FSM)

Total: ~140-180 LEs, ~1040 registers

4.7 2.1.7 Integration Guidelines

4.7.1 When to Use Generic Blocks

Use directly when: - Building custom data pipelines - Data width conversion without AXI4 protocol - Simple valid/ready streaming interfaces

Use full converters when: - Need AXI4 channel management (AW, W, B, AR, R) - Burst length adjustment required - ID tracking needed

4.7.2 Composition Example

```
// Full AXI4 write path composition
axi4_dwidth_converter_wr #(.S_DATA_WIDTH(64), .M_DATA_WIDTH(512)) u_wr
(
    // This module internally instantiates:
    // 1. Address phase skid buffer
    // 2. axi_data_upsize for write data
    // 3. Response path passthrough
    // 4. Burst length adjustment logic
);
```

Next: [axi_data_upsize Module](#)

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5 2.2 axi_data_upsize Module

The **axi_data_upsize** module accumulates N narrow beats into 1 wide beat. It is the core building block for narrow-to-wide data width conversion.

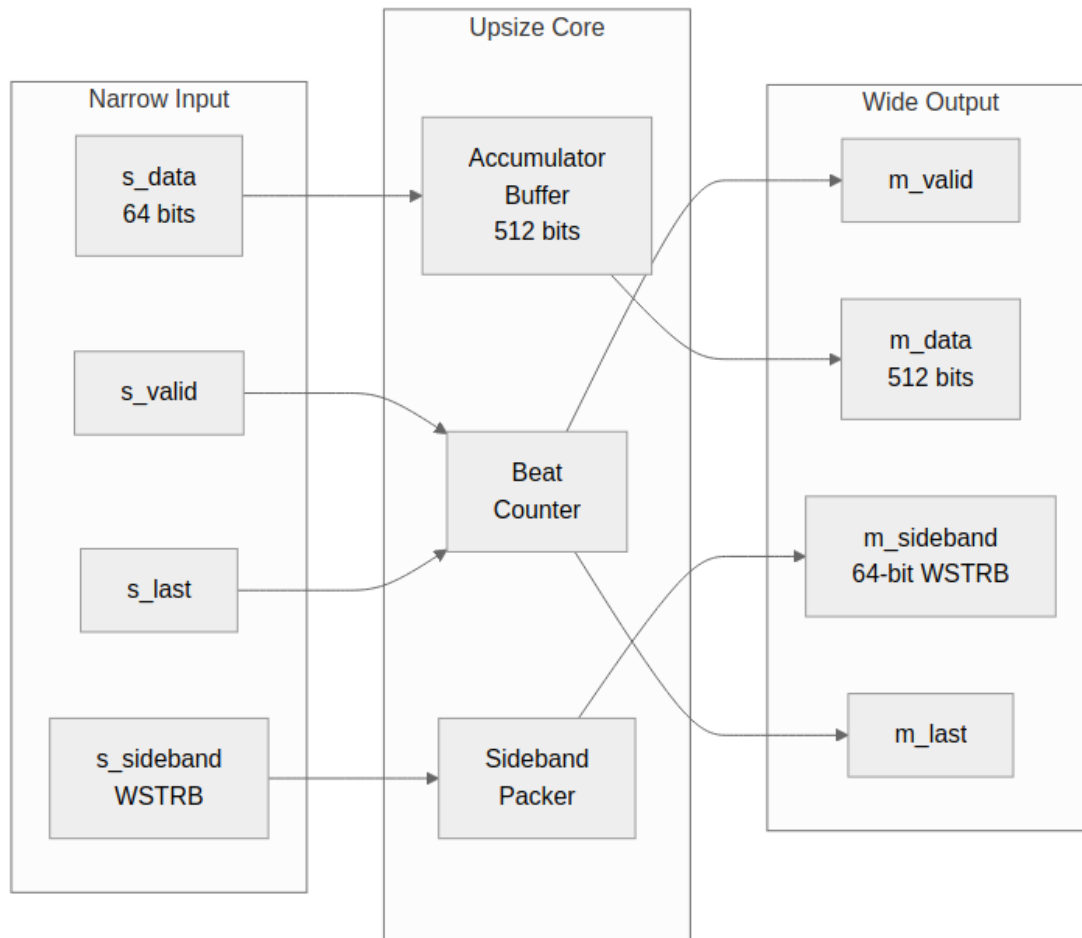
5.1 2.2.1 Purpose and Function

The upsize module does four things:

1. **Data Accumulation:** Collects N narrow data beats into accumulator buffer
2. **Sideband Packing:** Concatenates or ORs sideband signals (WSTRB, etc.)
3. **Flow Control:** Manages valid/ready handshaking with single-cycle latency
4. **LAST Tracking:** Detects input LAST to generate output LAST

5.2 2.2.2 Block Diagram

5.2.1 Figure 2.2: axi_data_upsize Architecture



axi_data_upsize Architecture

5.3 2.2.3 Interface Specification

5.3.1 Parameters

Table 2.4: *axi_data_upsize* Parameters

Parameter	Type	Default	Description
NARROW_WIDTH	int	64	Input data width (bits)
WIDE_WIDTH	int	512	Output data width (bits)
NARROW_SB_WIDTH	int	8	Input sideband width (bits)
WIDE_SB_WIDTH	int	64	Output sideband width
SB_OR_MODE	bit	0	0=concatenate, 1=OR sidebands
USE_LAST	bit	1	Enable LAST signal tracking

5.3.2 Ports

```
module axi_data_upsize #(
    parameter int NARROW_WIDTH    = 64,
    parameter int WIDE_WIDTH      = 512,
    parameter int NARROW_SB_WIDTH = 8,
    parameter int WIDE_SB_WIDTH   = 64,
    parameter bit SB_OR_MODE      = 0,
    parameter bit USE_LAST        = 1
) (
    input logic                clk,
    input logic                rst_n,

    // Narrow input interface
    input logic                s_valid,
    output logic               s_ready,
    input logic [NARROW_WIDTH-1:0] s_data,
    input logic [NARROW_SB_WIDTH-1:0] s_sideband,
    input logic                s_last,

    // Wide output interface
```

```
output logic m_valid,
input logic m_ready,
output logic [WIDE_WIDTH-1:0] m_data,
output logic [WIDE_SB_WIDTH-1:0] m_sideband,
output logic m_last
);
```

5.4 2.2.4 Operation

5.4.1 Accumulation Cycle

Cycle 0: s_data[0] → buffer[63:0], count = 0 → 1

Cycle 1: s_data[1] → buffer[127:64], count = 1 → 2

...

Cycle 7: s_data[7] → buffer[511:448], count = 7 → 0, m_valid = 1

Cycle 8: m_ready handshake, output complete

5.4.2 Early LAST Handling

If s_last arrives before buffer is full:

Cycle 0: s_data[0] → buffer[63:0], count = 0 → 1

Cycle 1: s_data[1] + s_last → buffer[127:64], count = 1 → 0

m_valid = 1, m_last = 1

Remaining bytes = don't care (masked by WSTRB)

5.4.3 State Machine

IDLE (count=0):

- s_valid=1 → load beat, increment count
- count < RATIO-1 → stay in IDLE
- count = RATIO-1 OR s_last → OUTPUT

OUTPUT (m_valid=1):

- m_ready=1 → clear buffer, → IDLE
- m_ready=0 → hold output

5.5 2.2.5 Sideband Handling

5.5.1 Concatenate Mode (SB_OR_MODE=0)

Used for WSTRB packing:

```
// Pack narrow sidebands into wide sideband
always_ff @(posedge clk) begin
    if (s_valid && s_ready) begin
        r_sideband[r_count * NARROW_SB_WIDTH +: NARROW_SB_WIDTH] <=
s_sideband;
    end
end
```

Example: 8 beats of 8-bit WSTRB to 64-bit WSTRB

```
Beat 0: WSTRB = 0xFF → output[7:0] = 0xFF
Beat 1: WSTRB = 0xF0 → output[15:8] = 0xF0
Beat 2: WSTRB = 0x0F → output[23:16] = 0x0F
...
Beat 7: WSTRB = 0xAA → output[63:56] = 0xAA
Final: output = 0xAA..._0F_F0_FF
```

5.5.2 OR Mode (SB_OR_MODE=1)

Used for error flag aggregation:

```
// OR narrow sidebands together
always_ff @(posedge clk) begin
    if (s_valid && s_ready) begin
        if (r_count == 0)
            r_sideband <= {{(WIDE_SB_WIDTH-NARROW_SB_WIDTH){1'b0}},
s_sideband};
        else
            r_sideband <= r_sideband | s_sideband;
    end
end
```

Example: Any error in burst propagates

```
Beat 0: error = 0 → output = 0
Beat 1: error = 0 → output = 0
Beat 2: error = 1 → output = 1 (error detected)
Beat 3: error = 0 → output = 1 (remains set)
...
Final: output = 1 (error occurred in burst)
```

5.6 2.2.6 Implementation

5.6.1 Core Logic

```
// Beat counter
logic [$clog2(RATIO)-1:0] r_count;

// Accumulator buffer
logic [WIDE_WIDTH-1:0] r_data;
logic [WIDE_SB_WIDTH-1:0] r_sideband;
logic r_last;

// Output valid when buffer full or early LAST
logic w_output_valid;
assign w_output_valid = (r_count == RATIO - 1) || r_last;

// Ready when not outputting or downstream ready
assign s_ready = !w_output_valid || m_ready;

// Main accumulation logic
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_count <= '0;
        r_data <= '0;
        r_sideband <= '0;
        r_last <= 1'b0;
    end else if (s_valid && s_ready) begin
        // Pack data into buffer
        r_data[r_count * NARROW_WIDTH +: NARROW_WIDTH] <= s_data;

        // Handle sideband based on mode
        if (SB_OR_MODE)
            r_sideband <= (r_count == 0) ? s_sideband : (r_sideband |
s_sideband);
        else
            r_sideband[r_count * NARROW_SB_WIDTH +: NARROW_SB_WIDTH]
<= s_sideband;

        // Track LAST
        r_last <= s_last;

        // Update counter
        if (s_last || r_count == RATIO - 1)
            r_count <= '0;
        else
            r_count <= r_count + 1'b1;
    end
```

```

        end else if (m_valid && m_ready) begin
            r_last <= 1'b0;
        end
    end

    // Output assignments
    assign m_valid = w_output_valid;
    assign m_data = r_data;
    assign m_sideband = r_sideband;
    assign m_last = r_last;

```

5.7 2.2.7 Timing Characteristics

5.7.1 Latency

Table 2.5: Upsize Latency

Scenario	Latency
Full buffer (N beats)	N cycles
Early LAST (M beats)	M cycles
Output handshake	0-1 cycles

5.7.2 Throughput

100% throughput - no gaps required between input beats.

The accumulator accepts one beat per cycle, every cycle. When the output buffer completes its handshake, accumulation of the next wide beat starts immediately.

5.7.3 Critical Paths

Typical critical paths: - s_data → accumulator buffer → m_data - r_count → comparison → s_ready

Timing closure: The module is designed for single-cycle operation, with combinatorial paths only within registered stages.

5.8 2.2.8 Resource Utilization

5.8.1 Typical Resources (64-bit to 512-bit)

Accumulator buffer: 512 flip-flops
Sideband buffer: 64 flip-flops (WSTRB)
Beat counter: 3 flip-flops
Control logic: ~20 flip-flops
~50-70 LUTs

Total: ~600 flip-flops, ~50-70 LUTs

5.8.2 Scaling

Table 2.6: Upsize Resource Scaling

Configuration	Registers	LUTs
32 → 128 (4:1)	~170	~30
64 → 256 (4:1)	~330	~40
64 → 512 (8:1)	~600	~60
128 → 1024 (8:1)	~1150	~80

5.9 2.2.9 Usage Example

5.9.1 64-bit to 512-bit Write Data

```
axi_data_upsize #(
    .NARROW_WIDTH(64),
    .WIDE_WIDTH(512),
    .NARROW_SB_WIDTH(8),    //WSTRB
    .WIDE_SB_WIDTH(64),
    .SB_OR_MODE(0),        //ConcatenateWSTRB
    .USE_LAST(1)
) u_wdata_upsize (
    .clk          (aclk),
    .rst_n        (aresetn),
    .s_valid      (s_wvalid),
    .s_ready      (s_wready),
    .s_data       (s_wdata),
    .s_sideband   (s_wstrb),
    .s_last       (s_wlast),
    .m_valid      (m_wvalid),
    .m_ready      (m_wready),
    .m_data       (m_wdata),
    .m_sideband   (m_wstrb),
    .m_last       (m_wlast)
);
```

Next: [axi_data_dsize Module](#)

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6 2.3 axi_data_dsize Module

The **axi_data_dsize** module splits 1 wide beat into N narrow beats. It supports both single-buffer (80% throughput) and dual-buffer (100% throughput) modes.

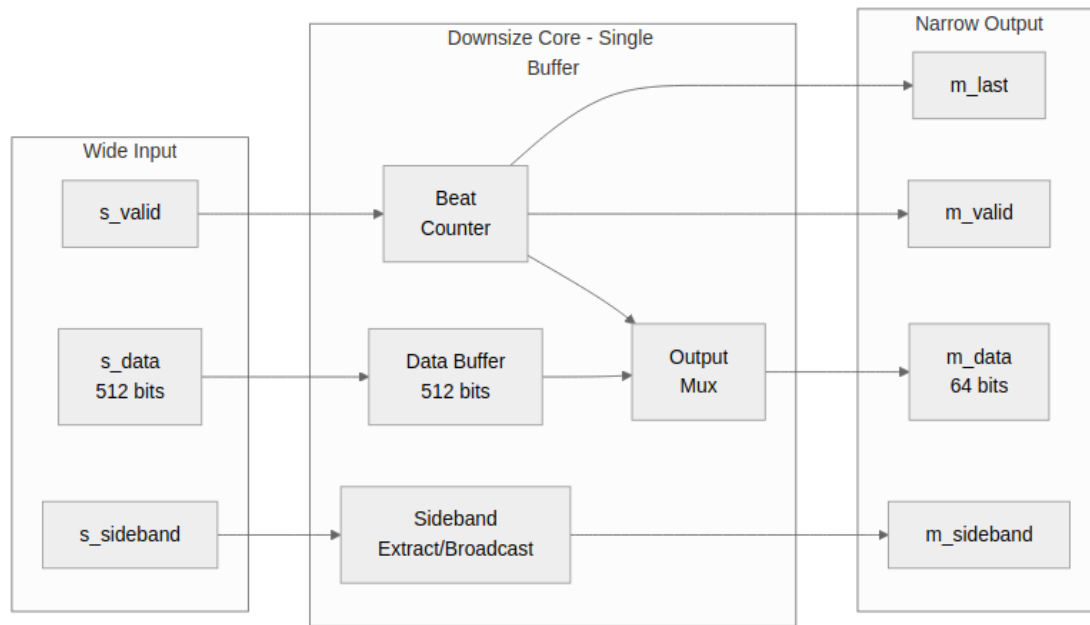
6.1 2.3.1 Purpose and Function

The downsize module does four things:

1. **Data Splitting:** Extracts N narrow beats from one wide beat
2. **Sideband Extraction:** Slices or broadcasts sideband signals
3. **Dual-Buffer Mode:** Optional ping-pong buffering for 100% throughput
4. **Burst Tracking:** Optional LAST signal generation based on burst length

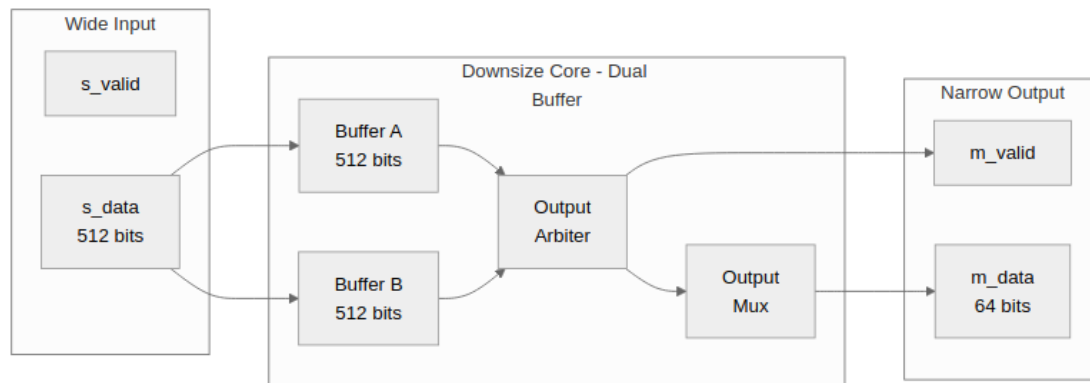
6.2 2.3.2 Block Diagram

6.2.1 Figure 2.3: axi_data_dnsiz Single-Buffer Architecture



axi_data_dnsiz Single Buffer

6.2.2 Figure 2.4: axi_data_dnsiz Dual-Buffer Architecture



axi_data_dnsiz Dual Buffer

6.3 2.3.3 Interface Specification

6.3.1 Parameters

Table 2.7: *axi_data_dnsiz*e Parameters

Parameter	Type	Default	Description
WIDE_WIDTH	int	512	Input data width (bits)
NARROW_WIDTH	int	64	Output data width (bits)
WIDE_SB_WIDTH	int	2	Input sideband width
NARROW_SB_WIDTH	int	2	Output sideband width
SB_BROADCAST	bit	1	0=slice, 1=broadcast sidebands
DUAL_BUFFER	bit	0	0=single, 1=dual buffer mode
USE_BURST_TRACKER	bit	0	Enable burst-aware LAST generation
BURST_LEN_WIDTH	int	8	Width of burst length input

6.3.2 Ports

```
module axi_data_dnsiz #(
    parameter int WIDE_WIDTH      = 512,
    parameter int NARROW_WIDTH    = 64,
    parameter int WIDE_SB_WIDTH   = 2,
    parameter int NARROW_SB_WIDTH = 2,
    parameter bit SB_BROADCAST    = 1,
    parameter bit DUAL_BUFFER     = 0,
    parameter bit USE_BURST_TRACKER = 0,
    parameter int BURST_LEN_WIDTH = 8
) (
    input logic                clk,
```

```

input logic                                rst_n,

// Wide input interface
input logic                                s_valid,
output logic                               s_ready,
input logic [WIDE_WIDTH-1:0]              s_data,
input logic [WIDE_SB_WIDTH-1:0]           s_sideband,
input logic                               s_last,

// Burst length input (optional)
input logic [BURST_LEN_WIDTH-1:0] burst_len,

// Narrow output interface
output logic                               m_valid,
input logic                               m_ready,
output logic [NARROW_WIDTH-1:0]           m_data,
output logic [NARROW_SB_WIDTH-1:0] m_sideband,
output logic                               m_last
);

```

6.4 2.3.4 Single-Buffer Mode Operation

6.4.1 Split Cycle

Cycle 0: s_data loaded → buffer, s_ready = 0
Cycle 1: m_data = buffer[63:0], count = 0, m_valid = 1
Cycle 2: m_data = buffer[127:64], count = 1
...
Cycle 8: m_data = buffer[511:448], count = 7, m_last possible
Cycle 9: s_ready = 1, gap cycle (80% throughput loss)

6.4.2 State Machine

IDLE:

- s_valid=1 → load buffer → SPLITTING

SPLITTING:

- Output beats 0 to RATIO-1
- m_ready=1 → increment count
- count=RATIO-1 AND m_ready → IDLE

6.4.3 Throughput Analysis

Why 80%?

For ratio N, each wide beat costs: - N cycles outputting narrow beats - 1 cycle loading next wide beat

Throughput = $N / (N + 1)$

Table 2.8: Single-Buffer Throughput by Ratio

Ratio	Cycles Active	Cycles Total	Throughput
2:1	2	3	66.7%
4:1	4	5	80.0%
8:1	8	9	88.9%
16:1	16	17	94.1%

6.5 2.3.5 Dual-Buffer Mode Operation

6.5.1 Ping-Pong Operation

Buffer A	Buffer B	Output
-----	-----	-----
Load beat 0	(empty)	(idle)
Outputting 0	Load beat 1	beat 0[0]
Outputting 0	Outputting 1	beat 0[1]
...	...	...
Outputting 0	Outputting 1	beat 0[N-1]
Load beat 2	Outputting 1	beat 1[0]
Outputting 2	Outputting 1	beat 1[1]
...		

6.5.2 State Machine (Dual)

Buffer A State: LOADING | OUTPUTTING | DONE
Buffer B State: LOADING | OUTPUTTING | DONE

Arbiter selects active output buffer
When output complete, swap buffers

6.5.3 100% Throughput

Dual-buffer achieves 100% throughput because loading and draining overlap: while one buffer outputs, the other loads. No gap cycles required.

Trade-off: 2x register resources

6.6 2.3.6 Sideband Handling

6.6.1 Slice Mode (SB_BROADCAST=0)

Used for WSTRB extraction:

```
// Extract sideband slice per beat  
assign m_sideband = r_sideband[r_count * NARROW_SB_WIDTH +:  
NARROW_SB_WIDTH];
```

Example: 64-bit WSTRB to 8-bit WSTRB

Input: 0xAA_BB_CC_DD_EE_FF_00_11

Beat 0: output WSTRB = 0x11 (bits [7:0])
Beat 1: output WSTRB = 0x00 (bits [15:8])
Beat 2: output WSTRB = 0xFF (bits [23:16])
...
Beat 7: output WSTRB = 0xAA (bits [63:56])

6.6.2 Broadcast Mode (SB_BROADCAST=1)

Used for RRESP:

```
// Broadcast same sideband to all beats  
assign m_sideband = r_sideband[NARROW_SB_WIDTH-1:0];
```

Example: RRESP = OKAY for all beats

Input: RRESP = 2'b00 (OKAY)

Beat 0: output RRESP = 2'b00
Beat 1: output RRESP = 2'b00
...
Beat 7: output RRESP = 2'b00

6.7 2.3.7 Burst Tracking

6.7.1 Purpose

When USE_BURST_TRACKER=1, the module generates correct m_last based on AXI4 burst length instead of relying on input s_last.

6.7.2 Operation

```
// Track narrow beats across burst
logic [BURST_LEN_WIDTH+RATIO_LOG2-1:0] r_burst_beats_remaining;

// Initialize on first beat
if (first_beat)
    r_burst_beats_remaining <= (burst_len + 1) * RATIO - 1;

// Decrement on each output
if (m_valid && m_ready)
    r_burst_beats_remaining <= r_burst_beats_remaining - 1;

// Generate LAST
assign m_last = (r_burst_beats_remaining == 0);
```

Example: ARLEN=3 (4 beats), ratio 8:1

Total narrow beats = $4 * 8 = 32$

Beat 0-30: m_last = 0

Beat 31: m_last = 1

6.8 2.3.8 Implementation

6.8.1 Single-Buffer Core Logic

```
// Beat counter
logic [$clog2(RATIO)-1:0] r_count;
logic r_active;

// Data buffer
logic [WIDE_WIDTH-1:0] r_data;
logic [WIDE_SB_WIDTH-1:0] r_sideband;
logic r_last_wide;

// Load/output control
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_active <= 1'b0;
        r_count <= '0;
    end else begin
        if (!r_active && s_valid) begin
            // Load new wide beat
            r_data <= s_data;
            r_sideband <= s_sideband;
            r_last_wide <= s_last;
            r_active <= 1'b1;
            r_count <= '0;
        end else if (r_active && m_ready) begin
            if (r_count == RATIO - 1) begin
                r_active <= 1'b0; // Done with this beat
            end else begin
                r_count <= r_count + 1'b1;
            end
        end
    end
end

// Output data slice
assign m_data = r_data[r_count * NARROW_WIDTH +: NARROW_WIDTH];

// Sideband (slice or broadcast)
assign m_sideband = SB_BROADCAST ?
    r_sideband[NARROW_SB_WIDTH-1:0] :
    r_sideband[r_count * NARROW_SB_WIDTH +: NARROW_SB_WIDTH];

// Control signals
assign m_valid = r_active;
```

```
assign s_ready = !r_active;  
assign m_last = r_last_wide && (r_count == RATIO - 1);
```

6.9 2.3.9 Resource Utilization

6.9.1 Single-Buffer (512-bit to 64-bit)

Data buffer: 512 flip-flops
Sideband buffer: 64 flip-flops
Beat counter: 3 flip-flops
Control logic: ~10 flip-flops
~30-50 LUTs

Total: ~590 flip-flops, ~30-50 LUTs

6.9.2 Dual-Buffer (512-bit to 64-bit)

Data buffers (2x): 1024 flip-flops
Sideband buffers: 128 flip-flops
Beat counters (2x): 6 flip-flops
Control logic: ~30 flip-flops
Ping-pong FSM: ~50 LUTs

Total: ~1190 flip-flops, ~80-100 LUTs

6.9.3 Comparison

Table 2.9: Resource Comparison

Mode	Registers	LUTs	Throughput
Single	590	40	80%
Dual	1190	90	100%

Decision Guide: - Area-constrained: use single buffer, accept 80% throughput - Performance-critical: use dual buffer, accept 2x resources

6.10 2.3.10 Usage Example

6.10.1 512-bit to 64-bit Read Data (High Performance)

```
axi_data_dsize #(
    .WIDE_WIDTH(512),
    .NARROW_WIDTH(64),
    .WIDE_SB_WIDTH(2),           // RRESP
    .NARROW_SB_WIDTH(2),
    .SB_BROADCAST(1),           // Broadcast RRESP
    .DUAL_BUFFER(1),             // 100% throughput
    .USE_BURST_TRACKER(1),       // Generate RLAST
    .BURST_LEN_WIDTH(8)
) u_rdata_dsize (
    .clk          (aclk),
    .rst_n        (aresetn),
    .s_valid      (s_rvalid),
    .s_ready      (s_rready),
    .s_data       (s_rdata),
    .s_sideband   (s_rresp),
    .s_last       (s_rlast),
    .burst_len    (ar_len),
    .m_valid      (m_rvalid),
    .m_ready      (m_rready),
    .m_data       (m_rdata),
    .m_sideband   (m_rresp),
    .m_last       (m_rlast)
);
```

Next: [Dual-Buffer Architecture](#)

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7 2.4 Dual-Buffer Architecture

The dual-buffer mode for **axi_data_dsize** achieves 100% throughput by using ping-pong buffering to eliminate the gap cycle between wide beat loads.

7.1 2.4.1 Problem Statement

7.1.1 Single-Buffer Limitation

In single-buffer mode, one gap cycle is unavoidable between draining one wide beat and starting the next:

Cycle N: Output last narrow beat of wide beat A

Cycle N+1: Load wide beat B (gap - no output)

Cycle N+2: Output first narrow beat of wide beat B

This gap cycle reduces throughput to $N/(N+1)$, or approximately 80-90% depending on the width ratio.

7.1.2 High-Performance Requirements

Some paths can't absorb that loss: - DDR memory controllers with sustained bandwidth - DMA engines with continuous data flows - Real-time video/audio processing

7.2 2.4.2 Solution: Ping-Pong Buffering

7.2.1 Concept

The fix is two buffers that alternate between loading and outputting:

Time:	0	1	2	3	4	5	6	7	8	9	10	11
Buffer A:	LD	00	01	02	03	04	05	06	07	LD	00	01
Buffer B:		LD	--	--	--	--	--	--	--	LD	--	--
Output:	--	A0	A1	A2	A3	A4	A5	A6	A7	B0	B1	B2

LD = Loading, On = Outputting beat n

While buffer A outputs its 8 narrow beats, buffer B loads the next wide beat. When buffer A completes, buffer B immediately starts outputting while buffer A loads.

7.2.2 Figure 2.5: Dual-Buffer Ping-Pong Operation

Dual-Buffer Operation

Dual-Buffer Operation

7.3 2.4.3 Implementation

7.3.1 Buffer State Machine

Each buffer has three states:

```
typedef enum logic [1:0] {  
    BUF_IDLE      = 2'b00, // Empty, ready to load  
    BUF_LOADED    = 2'b01, // Full, waiting to output  
    BUF_OUTPUTTING = 2'b10 // Active output  
} buf_state_t;
```

```
buf_state_t r_buf_a_state, r_buf_b_state;
```

7.3.2 Output Arbiter

```
// Select which buffer outputs  
logic w_output_sel; // 0 = buffer A, 1 = buffer B
```

```
always_comb begin  
    if (r_buf_a_state == BUF_OUTPUTTING)  
        w_output_sel = 1'b0;  
    else if (r_buf_b_state == BUF_OUTPUTTING)  
        w_output_sel = 1'b1;  
    else if (r_buf_a_state == BUF_LOADED)  
        w_output_sel = 1'b0; // A ready first  
    else if (r_buf_b_state == BUF_LOADED)  
        w_output_sel = 1'b1; // B ready  
    else  
        w_output_sel = 1'b0; // Default  
end
```

7.3.3 Load Controller

```
// Determine which buffer can accept new data  
logic w_load_sel; // 0 = buffer A, 1 = buffer B
```

```
always_comb begin  
    if (r_buf_a_state == BUF_IDLE)  
        w_load_sel = 1'b0;  
    else if (r_buf_b_state == BUF_IDLE)  
        w_load_sel = 1'b1;  
    else  
        w_load_sel = 1'b0; // No buffer available  
end
```

```
assign s_ready = (r_buf_a_state == BUF_IDLE) || (r_buf_b_state ==  
BUF_IDLE);
```

7.3.4 Buffer State Transitions

```
// Buffer A state machine
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_buf_a_state <= BUF_IDLE;
    end else begin
        case (r_buf_a_state)
            BUF_IDLE: begin
                if (s_valid && s_ready && !w_load_sel) begin
                    r_buf_a_data <= s_data;
                    r_buf_a_state <= BUF_LOADED;
                end
            end
            BUF_LOADED: begin
                if (!w_output_sel || r_buf_b_state != BUF_OUTPUTTING)
                    r_buf_a_state <= BUF_OUTPUTTING;
            end
            BUF_OUTPUTTING: begin
                if (m_ready && r_buf_a_count == RATIO - 1)
                    r_buf_a_state <= BUF_IDLE;
            end
        endcase
    end
end

// Buffer B follows same pattern with opposite selection
```

7.4 2.4.4 Timing Analysis

7.4.1 Continuous Operation

With dual buffering, the output is continuous:

Table 2.10: Dual-Buffer Cycle Timing

Cycle	Buffer A	Buffer B	Output
0	Load W0	Idle	-
1	Output W0[0]	Load W1	W0[0]
2	Output W0[1]	Wait	W0[1]
...	...	...	...
8	Output W0[7]	Wait	W0[7]
9	Load W2	Output W1[0]	W1[0]
10	Wait	Output W1[1]	W1[1]

7.4.2 Latency

Table 2.11: Dual-Buffer Latency

Metric	Value
First beat latency	1 cycle (load)
Sustained throughput	100%
Buffer switch latency	0 cycles

7.5 2.4.5 Edge Cases

7.5.1 Empty Pipeline Start

When starting from empty state: 1. First wide beat loads into buffer A 2. Buffer A starts outputting 3. Buffer B loads next wide beat 4. Continuous operation begins

7.5.2 Pipeline Drain

When input stops (last beat processed): 1. Currently outputting buffer completes 2. Other buffer (if loaded) takes over 3. Pipeline drains to empty

7.5.3 Backpressure Handling

If `m_ready` goes low: 1. Output stalls on current beat 2. Loading continues if buffer available 3. If both buffers full, `s_ready` goes low 4. Resumes when `m_ready` returns high

7.6 2.4.6 Resource Comparison

7.6.1 Resource Breakdown

Table 2.12: Resource Breakdown

Component	Single Buffer	Dual Buffer
Data registers	512 bits	1024 bits
Sideband registers	64 bits	128 bits
Beat counters	3 bits	6 bits
State machines	1	2 + arbiter
Control logic	~40 LEs	~90 LEs

7.6.2 Trade-off Summary

Table 2.13: Mode Trade-offs

Mode	Area	Throughput	Use Case
Single	1x	80-90%	Area-constrained, bursty traffic
Dual	2x	100%	Performance-critical, sustained traffic

7.7 2.4.7 Configuration Guidelines

7.7.1 When to Use Dual Buffer

Recommended: - DDR/HBM memory interfaces - DMA engines with sustained transfers - Video/audio streaming paths - Any path where 10-20% throughput loss is unacceptable

Not Recommended: - Area-constrained designs - Bursty traffic with gaps - Control paths (low bandwidth) - When latency is more critical than throughput

7.7.2 Integration Example

```
// High-performance read path for DDR controller
axi_data_dsize #(
    .WIDE_WIDTH(512),
    .NARROW_WIDTH(64),
    .DUAL_BUFFER(1),           // Enable dual buffer
    .USE_BURST_TRACKER(1),
    .SB_BROADCAST(1)
) u_rdata_dsize (
    // ... connections
);

// Low-bandwidth control path (save area)
axi_data_dsize #(
    .WIDE_WIDTH(128),
    .NARROW_WIDTH(32),
    .DUAL_BUFFER(0),          // Single buffer OK
    .USE_BURST_TRACKER(0),
    .SB_BROADCAST(1)
) u_ctrl_dsize (
    // ... connections
);
```

Next: [axi4_dwidth_converter_wr](#)

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8 2.5 axi4_dwidth_converter_wr

The **axi4_dwidth_converter_wr** module is the complete AXI4 write path — AW, W, and B channels with burst length adjustment for the new width.

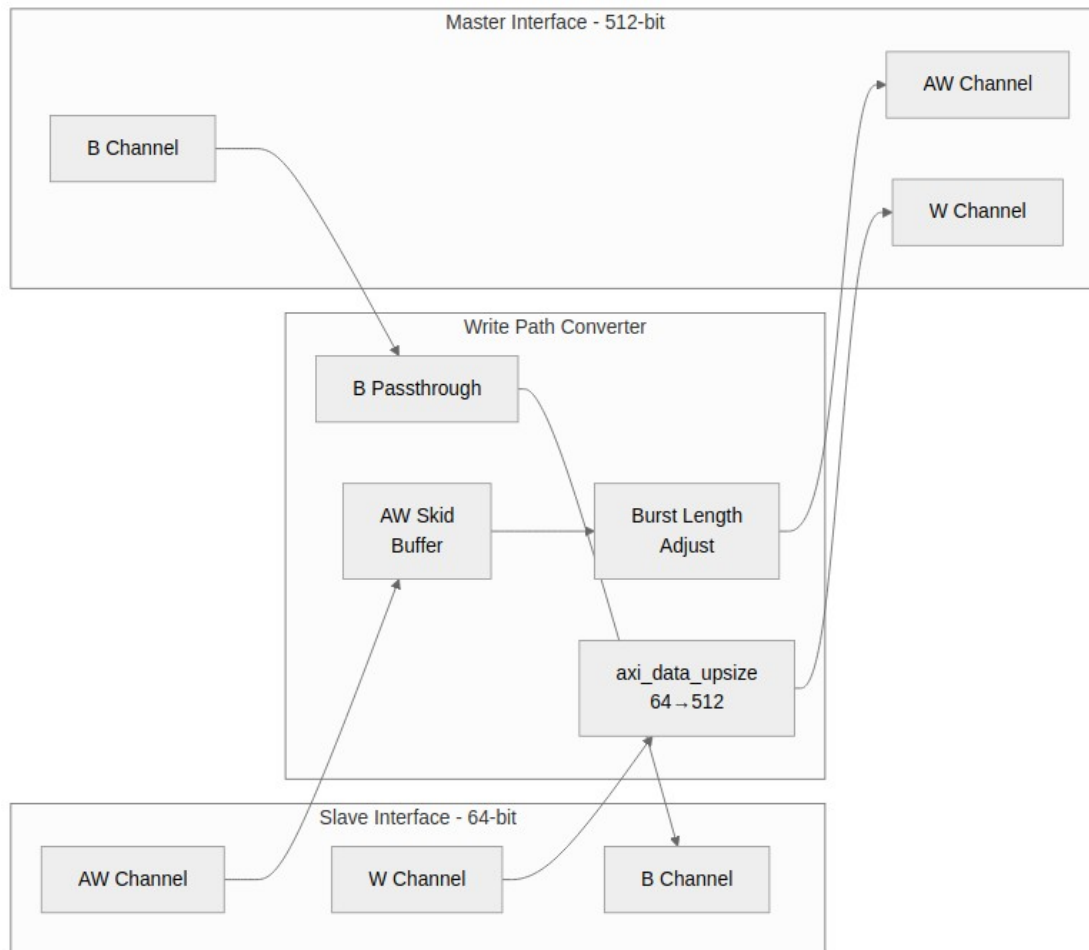
8.1 2.5.1 Purpose and Function

The write converter combines the generic `axi_data_upsize` with AXI4 protocol handling:

1. **Address Channel (AW):** Passes through with burst length adjustment
2. **Write Data Channel (W):** Uses `axi_data_upsize` for data packing
3. **Response Channel (B):** Passes through unchanged
4. **Burst Length Adjustment:** Converts AWLEN based on width ratio

8.2 2.5.2 Block Diagram

8.2.1 Figure 2.6: Write Converter Architecture



Write Converter Architecture

8.3 2.5.3 Interface Specification

8.3.1 Parameters

Table 2.14: Write Converter Parameters

Parameter	Type	Default	Description
S_DATA_WIDTH	int	64	Slave-side (narrow) data width
M_DATA_WIDTH	int	512	Master-side (wide) data width
ADDR_WIDTH	int	64	Address width
ID_WIDTH	int	4	Transaction ID width
SKID_DEPTH	int	2	Pipeline buffer depth

8.3.2 Ports

```
module axi4_dwidth_converter_wr #(
    parameter int S_DATA_WIDTH = 64,
    parameter int M_DATA_WIDTH = 512,
    parameter int ADDR_WIDTH   = 64,
    parameter int ID_WIDTH     = 4,
    parameter int SKID_DEPTH   = 2
) (
    input  logic clk,
    input  logic rst_n,

    // Slave interface (narrow, from master)
    input  logic          s_awvalid,
    output logic          s_awready,
    input  logic [ADDR_WIDTH-1:0] s_awaddr,
    input  logic [7:0]          s_awlen,
    input  logic [2:0]          s_awsz,
    input  logic [1:0]          s_awburst,
    input  logic [ID_WIDTH-1:0] s_awid,
    // ... other AW signals

    input  logic          s_wvalid,
    output logic          s_wready,
    input  logic [S_DATA_WIDTH-1:0] s_wdata,
```

```

input  logic [S_DATA_WIDTH/8-1:0] s_wstrb,
input  logic                                     s_wlast,

output logic                                     s_bvalid,
input  logic                                     s_bready,
output logic [ID_WIDTH-1:0]             s_bid,
output logic [1:0]                       s_bresp,

// Master interface (wide, to slave)
output logic                                     m_awvalid,
input  logic                                     m_awready,
output logic [ADDR_WIDTH-1:0]             m_awaddr,
output logic [7:0]                         m_awlen,
output logic [2:0]                         m_awsz,
output logic [1:0]                         m_awburst,
output logic [ID_WIDTH-1:0]             m_awid,
// ... other AW signals

output logic                                     m_wvalid,
input  logic                                     m_wready,
output logic [M_DATA_WIDTH-1:0]         m_wdata,
output logic [M_DATA_WIDTH/8-1:0]       m_wstrb,
output logic                                     m_wlast,

input  logic                                     m_bvalid,
output logic                                     m_bready,
input  logic [ID_WIDTH-1:0]             m_bid,
input  logic [1:0]                       m_bresp
);

```

8.4 2.5.4 Burst Length Conversion

8.4.1 Ratio Calculation

```
localparam int RATIO = M_DATA_WIDTH / S_DATA_WIDTH;  
localparam int RATIO_LOG2 = $clog2(RATIO);
```

```
// New AWLEN = (original AWLEN + 1) / RATIO - 1  
// = (AWLEN + 1) >> RATIO_LOG2 - 1
```

8.4.2 Examples

Table 2.15: Burst Length Conversion Examples

S_DATA	M_DATA	Ratio	S_AWLE	S_beats	M_AWLE	M_beats
			N		N	
64	512	8	7	8	0	1
64	512	8	15	16	1	2
64	256	4	3	4	0	1
64	256	4	7	8	1	2

8.4.3 Non-Aligned Bursts

When the burst length is not a multiple of the ratio:

```
S_AWLEN = 5 (6 beats), RATIO = 8  
M_AWLEN = 0 (1 beat)
```

The 6 narrow beats pack into 1 wide beat.
Remaining 2 positions have WSTRB = 0 (no write).

8.5 2.5.5 Address Channel Handling

8.5.1 AW Passthrough with Adjustment

```
// Burst length adjustment
logic [7:0] w_adjusted_awlen;
assign w_adjusted_awlen = ((s_awlen + 1) >> RATIO_LOG2) - 1;

// Size adjustment (wider data = larger size)
logic [2:0] w_adjusted_awsz;
assign w_adjusted_awsz = s_awsz + RATIO_LOG2;

// Address alignment check
logic w_aligned;
assign w_aligned = (s_awaddr[RATIO_LOG2+2:0] == '0);
```

8.5.2 Skid Buffer for AW

```
axi_skid_buffer #(
    .DATA_WIDTH(AW_CHANNEL_WIDTH)
) u_aw_skid (
    .clk      (clk),
    .rst_n    (rst_n),
    .s_valid  (s_awvalid),
    .s_ready  (s_awready),
    .s_data   ({s_awid, s_awaddr, w_adjusted_awlen, ...}),
    .m_valid  (m_awvalid),
    .m_ready  (m_awready),
    .m_data   ({m_awid, m_awaddr, m_awlen, ...})
);
```

8.6 2.5.6 Write Data Channel

8.6.1 Upsize Instance

```
axi_data_upsize #(
    .NARROW_WIDTH(S_DATA_WIDTH),
    .WIDE_WIDTH(M_DATA_WIDTH),
    .NARROW_SB_WIDTH(S_DATA_WIDTH/8), // WSTRB
    .WIDE_SB_WIDTH(M_DATA_WIDTH/8),
    .SB_OR_MODE(0), // Concatenate WSTRB
    .USE_LAST(1)
) u_wdata_upsize (
    .clk      (clk),
    .rst_n    (rst_n),
    .s_valid  (s_wvalid),
    .s_ready  (s_wready),
    .s_data   (s_wdata),
    .s_sideband (s_wstrb),
    .s_last   (s_wlast),
    .m_valid  (m_wvalid),
    .m_ready  (m_wready),
    .m_data   (m_wdata),
    .m_sideband (m_wstrb),
    .m_last   (m_wlast)
);
```

8.7 2.5.7 Response Channel

8.7.1 B Channel Passthrough

The response channel passes through unchanged:

```
// Simple passthrough (or via skid buffer)  
assign s_bvalid = m_bvalid;  
assign m_bready = s_bready;  
assign s_bid    = m_bid;  
assign s_bresp  = m_bresp;
```

8.7.2 Response Ordering

Responses return in order because: - Single outstanding transaction per ID - Upsize doesn't reorder data - B response generated after all W beats accepted

8.8 2.5.8 AW/W Synchronization

8.8.1 Challenge

AXI4 allows AW to arrive before, with, or after W data. The converter must handle all cases:

1. **AW before W:** Normal pipelining
2. **W before AW:** Data buffered until AW arrives
3. **Interleaved:** Multiple transactions in flight

8.8.2 Solution

A FIFO carries the AW information the upsize logic needs:

```
// FIFO stores AWLEN for burst tracking
fifo_sync #(.WIDTH(8), .DEPTH(4)) u_aw_info_fifo (
    .clk      (clk),
    .rst_n    (rst_n),
    .wr_en    (s_awvalid && s_awready),
    .wr_data  (s_awlen),
    .rd_en    (m_wvalid && m_wready && m_wlast),
    .rd_data  (current_awlen),
    .full     (aw_fifo_full),
    .empty    (aw_fifo_empty)
);
```

8.9 2.5.9 Resource Utilization

8.9.1 Typical Resources (64→512, ID=4, ADDR=64)

AW skid buffer: ~200 flip-flops
W upsize: ~600 flip-flops, ~60 LUTs
B skid buffer: ~20 flip-flops
AW info FIFO: ~50 flip-flops
Control logic: ~100 LUTs

Total: ~870 flip-flops, ~160 LUTs

8.10 2.5.10 Timing Characteristics

8.10.1 Latency

Table 2.16: Write Converter Latency

Path	Latency
AW passthrough	1-2 cycles (skid)
W upsize	N cycles (accumulation)
B passthrough	1 cycle (skid)

8.10.2 Throughput

- AW channel: 1 transaction/cycle
- W channel: 100% (upsized is 100%)
- B channel: 1 response/cycle

8.11 2.5.11 Usage Example

```
axi4_dwidth_converter_wr #(
    .S_DATA_WIDTH(64),
    .M_DATA_WIDTH(512),
    .ADDR_WIDTH(64),
    .ID_WIDTH(4),
    .SKID_DEPTH(2)
) u_wr_converter (
    .clk      (aclk),
    .rst_n    (aresetn),

    // 64-bit slave interface (from CPU)
    .s_awvalid (cpu_awvalid),
    .s_awready (cpu_awready),
    .s_awaddr  (cpu_awaddr),
    .s_awlen   (cpu_awlen),
    // ... other s_* signals

    // 512-bit master interface (to DDR)
    .m_awvalid (ddr_awvalid),
    .m_awready (ddr_awready),
    .m_awaddr  (ddr_awaddr),
    .m_awlen   (ddr_awlen),
    // ... other m_* signals
);
```

Next: [axi4_dwidth_converter_rd](#)

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9 2.6 axi4_dwidth_converter_rd

The **axi4_dwidth_converter_rd** module is the complete AXI4 read path — AR and R channels with burst length adjustment and burst-aware RLAST generation.

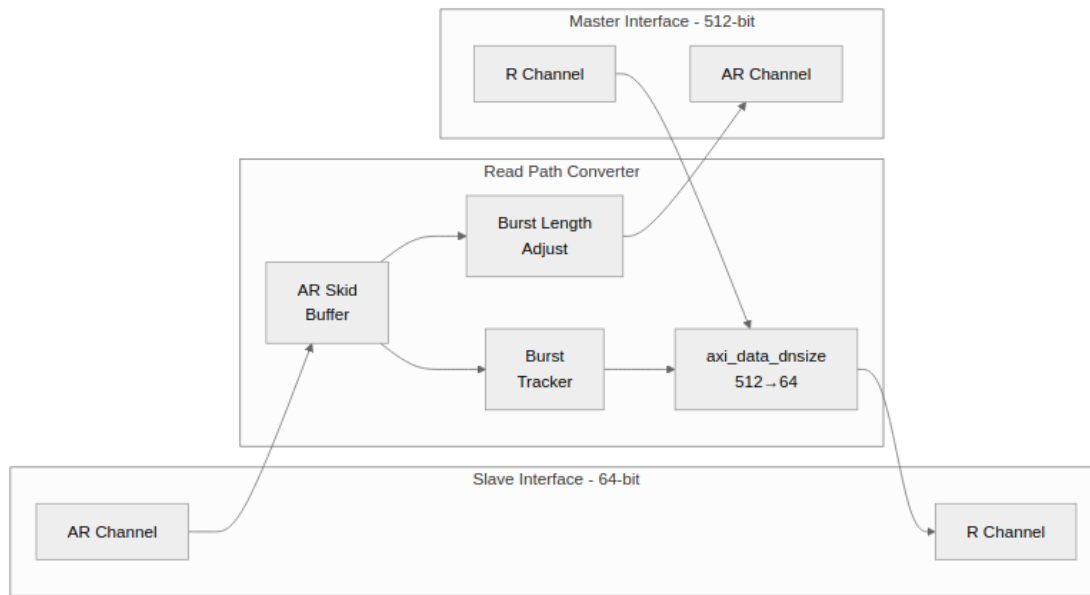
9.1 2.6.1 Purpose and Function

The read converter combines the generic `axi_data_dsize` with AXI4 protocol handling:

1. **Address Channel (AR):** Passes through with burst length adjustment
2. **Read Data Channel (R):** Uses `axi_data_dsize` for data splitting
3. **Burst Tracking:** Generates correct RLAST based on original ARLEN
4. **Response Broadcasting:** Propagates RRESP to all narrow beats

9.2 2.6.2 Block Diagram

9.2.1 Figure 2.7: Read Converter Architecture



Read Converter Architecture

9.3 2.6.3 Interface Specification

9.3.1 Parameters

Table 2.17: Read Converter Parameters

Parameter	Type	Default	Description
S_DATA_WIDTH	int	64	Slave-side (narrow) data width
M_DATA_WIDTH	int	512	Master-side (wide) data width
ADDR_WIDTH	int	64	Address width
ID_WIDTH	int	4	Transaction ID width
DUAL_BUFFER	bit	1	Enable dual-buffer for 100% throughput
SKID_DEPTH	int	2	Pipeline buffer depth

9.3.2 Ports

```
module axi4_dwidth_converter_rd #(
    parameter int S_DATA_WIDTH = 64,
    parameter int M_DATA_WIDTH = 512,
    parameter int ADDR_WIDTH   = 64,
    parameter int ID_WIDTH     = 4,
    parameter bit  DUAL_BUFFER = 1,
    parameter int SKID_DEPTH   = 2
) (
    input logic clk,
    input logic rst_n,

    // Slave interface (narrow, to master)
    input logic s_arvalid,
    output logic s_arready,
    input logic [ADDR_WIDTH-1:0] s_araddr,
    input logic [7:0] s_arlen,
    input logic [2:0] s_arsize,
    input logic [1:0] s_arburst,
```

```

input  logic [ID_WIDTH-1:0]      s_arid,

output logic                      s_rvalid,
input  logic                      s_rready,
output logic [S_DATA_WIDTH-1:0]  s_rdata,
output logic [ID_WIDTH-1:0]      s_rid,
output logic [1:0]               s_rresp,
output logic                      s_rlast,

// Master interface (wide, from slave)
output logic                      m_arvalid,
input  logic                      m_arready,
output logic [ADDR_WIDTH-1:0]    m_araddr,
output logic [7:0]               m_arlen,
output logic [2:0]               m_arsize,
output logic [1:0]               m_arburst,
output logic [ID_WIDTH-1:0]      m_arid,

input  logic                      m_rvalid,
output logic                      m_rready,
input  logic [M_DATA_WIDTH-1:0]  m_rdata,
input  logic [ID_WIDTH-1:0]      m_rid,
input  logic [1:0]               m_rresp,
input  logic                      m_rlast
);

```

9.4 2.6.4 Burst Length Conversion

9.4.1 Ratio Calculation

Same as write converter:

```
localparam int RATIO = M_DATA_WIDTH / S_DATA_WIDTH;  
localparam int RATIO_LOG2 = $clog2(RATIO);
```

```
// New ARLEN = (original ARLEN + 1) / RATIO - 1
```

9.4.2 Examples

Table 2.18: Read Burst Length Conversion

S_DATA	M_DATA	Ratio	S_ARLE	S_beats	M_ARLE	M_beats
			N		N	
64	512	8	7	8	0	1
64	512	8	15	16	1	2
64	512	8	31	32	3	4

9.5 2.6.5 Address Channel Handling

9.5.1 AR Passthrough with Adjustment

```
// Burst length adjustment
logic [7:0] w_adjusted_arlen;
assign w_adjusted_arlen = ((s_arlen + 1) >> RATIO_LOG2) - 1;

// Size adjustment
logic [2:0] w_adjusted_arsize;
assign w_adjusted_arsize = s_arsize + RATIO_LOG2;
```

9.5.2 AR Information FIFO

Store original ARLEN for RLAST generation:

```
// FIFO to track original burst length
fifo_sync #(.WIDTH(8+ID_WIDTH), .DEPTH(4)) u_ar_info_fifo (
    .clk      (clk),
    .rst_n    (rst_n),
    .wr_en     (s_arvalid && s_arready),
    .wr_data   ({s_arid, s_arlen}),
    .rd_en     (s_rvalid && s_rready && s_rlast),
    .rd_data   ({current_arid, current_arlen}),
    .full      (ar_fifo_full),
    .empty     (ar_fifo_empty)
);
```

9.6 2.6.6 Read Data Channel

9.6.1 Downsize Instance

```
axi_data_dnsz #(  
    .WIDE_WIDTH(M_DATA_WIDTH),  
    .NARROW_WIDTH(S_DATA_WIDTH),  
    .WIDE_SB_WIDTH(2),           // RRESP  
    .NARROW_SB_WIDTH(2),  
    .SB_BROADCAST(1),           // Broadcast RRESP  
    .DUAL_BUFFER(DUAL_BUFFER),  
    .USE_BURST_TRACKER(1),  
    .BURST_LEN_WIDTH(8)  
) u_rdata_dnsz (  
    .clk      (clk),  
    .rst_n    (rst_n),  
    .s_valid  (m_rvalid),  
    .s_ready  (m_rready),  
    .s_data   (m_rdata),  
    .s_sideband (m_rresp),  
    .s_last   (m_rlast),  
    .burst_len (current_arlen),  
    .m_valid   (s_rvalid),  
    .m_ready   (s_rready),  
    .m_data    (s_rdata),  
    .m_sideband (s_rresp),  
    .m_last    (s_rlast)  
);
```

9.7 2.6.7 RLAST Generation

9.7.1 Challenge

The wide interface generates RLAST based on M_ARLEN, but the narrow interface needs RLAST based on S_ARLEN:

Wide RLAST: Asserted on beat $(M_ARLEN + 1)$

Narrow RLAST: Asserted on beat $(S_ARLEN + 1) = (M_ARLEN + 1) * \text{RATIO}$

9.7.2 Solution: Burst Tracker

// Track narrow beats within burst

```
logic [15:0] r_beat_count;
```

```
logic [7:0] r_total_narrow_beats;
```

```
always_ff @(posedge clk or negedge rst_n) begin
```

```
    if (!rst_n) begin
```

```
        r_beat_count <= '0;
```

```
        r_total_narrow_beats <= '0;
```

```
    end else begin
```

```
        if (new_burst_start) begin
```

```
            r_beat_count <= '0;
```

```
            r_total_narrow_beats <= (current_arlen + 1) * RATIO - 1;
```

```
        end else if (s_rvalid && s_rready) begin
```

```
            r_beat_count <= r_beat_count + 1;
```

```
        end
```

```
    end
```

```
end
```

```
assign s_rlast = (r_beat_count == r_total_narrow_beats);
```

9.8 2.6.8 RID Handling

9.8.1 ID Passthrough

RID passes through unchanged:

```
// RID from wide interface propagates to all narrow beats  
assign s_rid = m_rid;
```

```
// Or use tracked ID from FIFO  
assign s_rid = current_arid;
```

9.9 2.6.9 Dual-Buffer Impact

9.9.1 Performance Comparison

Table 2.19: Read Converter Performance

Mode	Throughput	Latency	Resources
Single	80-90%	1 cycle	1x
Dual	100%	1 cycle	2x

9.9.2 When to Use Dual Buffer

Use dual buffer for: - DDR read paths with high bandwidth - Streaming read applications - DMA read operations

Use single buffer for: - Control register reads - Low-bandwidth paths - Area-constrained designs

9.10 2.6.10 Resource Utilization

9.10.1 Typical Resources (512→64, ID=4, Dual Buffer)

AR skid buffer: ~150 flip-flops
R downsize (dual): ~1200 flip-flops, ~100 LUTs
AR info FIFO: ~100 flip-flops
Burst tracker: ~30 flip-flops, ~20 LUTs
Control logic: ~80 LUTs

Total: ~1480 flip-flops, ~200 LUTs

9.10.2 Single Buffer Version

R downsize (single): ~600 flip-flops, ~50 LUTs

Total: ~880 flip-flops, ~120 LUTs

9.11 2.6.11 Timing Characteristics

9.11.1 Latency

Table 2.20: Read Converter Latency

Path	Latency
AR passthrough	1-2 cycles (skid)
First R beat	1 cycle (load buffer)
Subsequent R beats	1 beat/cycle

9.11.2 Throughput

- AR channel: 1 transaction/cycle
- R channel: 80% (single) or 100% (dual)

9.12 2.6.12 Usage Example

```
axi4_dwidth_converter_rd #(
    .S_DATA_WIDTH(64),
    .M_DATA_WIDTH(512),
    .ADDR_WIDTH(64),
    .ID_WIDTH(4),
    .DUAL_BUFFER(1),      // High-performance mode
    .SKID_DEPTH(2)
) u_rd_converter (
    .clk      (aclk),
    .rst_n    (aresetn),

    // 64-bit slave interface (to CPU)
    .s_arvalid (cpu_arvalid),
    .s_arready (cpu_arready),
    .s_araddr  (cpu_araddr),
    .s_arlen   (cpu_arlen),
    .s_rvalid  (cpu_rvalid),
    .s_rready  (cpu_rready),
    .s_rdata   (cpu_rdata),
    // ... other s_* signals

    // 512-bit master interface (from DDR)
    .m_arvalid (ddr_arvalid),
    .m_arready (ddr_arready),
    .m_araddr  (ddr_araddr),
    .m_arlen   (ddr_arlen),
    .m_rvalid  (ddr_rvalid),
    .m_rready  (ddr_rready),
    .m_rdata   (ddr_rdata),
    // ... other m_* signals
);
```

Next: [Chapter 3: Protocol Converter Blocks](#)

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10 3.1 Protocol Conversion Overview

Protocol converters bridge components that speak different bus protocols — the glue in any SoC that mixes AXI4, AXI4-Lite, and APB.

10.1 3.1.1 Available Converters

10.1.1 AXI4 to AXI4-Lite (Protocol Downgrade)

Table 3.1: AXI4 to AXI4-Lite Converters

Module	Function	Test Status
axi4_to_axil4_rd	Read burst decomposition	14/14 passing
axi4_to_axil4_wr	Write burst decomposition	14/14 passing
axi4_to_axil4	Full bidirectional wrapper	Composed

10.1.2 AXI4-Lite to AXI4 (Protocol Upgrade)

Table 3.2: AXI4-Lite to AXI4 Converters

Module	Function	Test Status
axil4_to_axi4_rd	Read protocol upgrade	7/7 passing
axil4_to_axi4_wr	Write protocol upgrade	7/7 passing
axil4_to_axi4	Full bidirectional wrapper	Composed

10.1.3 Other Protocol Converters

Table 3.3: Other Protocol Converters

Module	Function	Status
axi4_to_apb4_converter	Full AXI4-to-APB bridge	Production
peakrdl_to_cmdrsp	Register interface adapter	Production
uart_axil_bridge	UART to AXI4-Lite	Planned

10.2 3.1.2 Protocol Comparison

10.2.1 Figure 3.1: Protocol Feature Comparison

Protocol Comparison			
<i>Protocol Comparison</i>			

10.2.2 Feature Matrix

Table 3.4: Protocol Feature Comparison

Feature	AXI4	AXI4-Lite	APB
Channels	5 (AW, W, B, AR, R)	5 (simplified)	1 (combined)
Bursts	Up to 256 beats	Single beat only	Single beat
Out-of-order	Yes (ID-based)	No	No
Pipelining	Yes	Optional	No (2-phase)
Data widths	8-1024 bits	32/64 bits	8-32 bits

10.3 3.1.3 Conversion Strategies

10.3.1 AXI4 to AXI4-Lite

Challenge: Decompose multi-beat bursts into sequential single beats.

Strategy: 1. Accept AXI4 burst transaction 2. Issue N single-beat AXI4-Lite transactions 3. Aggregate responses 4. Return combined response to AXI4 master

Complexity: Medium (FSM-based decomposition)

10.3.2 AXI4-Lite to AXI4

Challenge: Add AXI4 burst signals with appropriate defaults.

Strategy: 1. Pass through all AXI4-Lite signals 2. Add default values for burst signals (LEN=0, SIZE=2, BURST=INCR) 3. Add default IDs (configurable)

Complexity: Very low (combinational only)

10.3.3 AXI4 to APB

Challenge: Bridge 5-channel AXI4 to 2-phase APB protocol.

Strategy: 1. Accept AXI4 transaction 2. Execute APB setup phase 3. Execute APB access phase 4. Return AXI4 response

Complexity: High (full protocol FSM)

10.4 3.1.4 Performance Characteristics

Table 3.5: Protocol Converter Performance

Converter	Single-Beat	Burst (N)	Area
axi4_to_axil4	0 cycles	2N cycles	~450 LUTs
axil4_to_axi4	0 cycles	N/A	~110 LUTs
axi4_to_apb4	3-5 cycles	(3-5)N cycles	~300 LUTs

10.4.1 Key Observations

1. **Zero-overhead upgrade:** AXI4-Lite to AXI4 is purely combinational
2. **Burst penalty:** AXI4 to AXI4-Lite doubles cycle count for bursts
3. **APB overhead:** 3-5 cycle minimum per APB transaction

10.5 3.1.5 Use Case Guidelines

10.5.1 When to Use AXI4 to AXI4-Lite

Use when: - CPU/DMA with burst support needs simple peripheral access - Want to simplify peripheral design (no burst handling) - Data widths match (no width conversion needed) - Burst performance is not critical

Avoid when: - High-bandwidth streaming data - Latency-critical paths - Many back-to-back bursts

10.5.2 When to Use AXI4-Lite to AXI4

Use when: - Legacy AXI4-Lite IP connects to AXI4 fabric - Designing simple peripheral for AXI4 system - Want zero-overhead protocol upgrade - Don't need burst capability

Avoid when: - Need actual burst support (use native AXI4) - Width conversion needed (use width converters)

10.5.3 When to Use AXI4 to APB

Use when: - AXI4 masters need APB peripheral access - Building CPU-to-peripheral bridges - Integrating legacy APB devices

Avoid when: - High-performance paths (APB is slow) - Streaming data (APB is sequential)

10.6 3.1.6 Integration Patterns

10.6.1 Pattern 1: CPU to Peripherals

CPU (AXI4) → AXI4-to-AXIL4 → AXIL4 Peripherals
→ AXI4-to-APB → APB Peripherals

10.6.2 Pattern 2: Simple IP in AXI4 Fabric

Simple IP (AXIL4) → AXIL4-to-AXI4 → AXI4 Crossbar

10.6.3 Pattern 3: Mixed System

CPU (AXI4) → AXI4 Crossbar → DDR (AXI4)
→ AXI4-to-AXIL4 → Config Regs (AXIL4)
→ AXI4-to-APB → UART, GPIO (APB)

Next: [AXI4 to AXI4-Lite](#)

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11 3.2 AXI4 to AXI4-Lite Converter

The **axi4_to_axil4** converter family decomposes AXI4 burst transactions into sequential AXI4-Lite single-beat transactions.

11.1 3.2.1 Module Organization

```
axi4_to_axil4.sv      # Full bidirectional wrapper
├─ axi4_to_axil4_rd.sv # Read path converter
└─ axi4_to_axil4_wr.sv # Write path converter
```

11.1.1 Design Philosophy

Read and write are separate modules, which buys: - Independent optimization - Selective instantiation (read-only, write-only, or both) - Simpler verification (test paths independently)

11.2 3.2.2 Read Path (axi4_to_axil4_rd)

11.2.1 Block Diagram

11.2.2 Figure 3.2: AXI4 to AXI4-Lite Read Path

AXI4 to AXIL4 Read

AXI4 to AXIL4 Read

11.2.3 Operation

Single-Beat (ARLEN=0):

Cycle 0: AR accepted (passthrough)
Cycle 1: R returned (passthrough)
Total: 0 extra cycles (pure passthrough)

Multi-Beat (ARLEN=N-1):

Cycle 0: AR[0] issued to AXIL4
Cycle 1: R[0] received, AR[1] issued
Cycle 2: R[1] received, AR[2] issued
...
Cycle 2N-1: R[N-1] received (RLAST)
Total: 2N cycles (1 cycle per AR + 1 cycle per R)

11.2.4 State Machine

```
typedef enum logic [1:0] {  
    IDLE           = 2'b00, // Wait for AR  
    DECOMPOSE      = 2'b01, // Issuing single beats  
    WAIT_R         = 2'b10  // Wait for last R  
} rd_state_t;
```

11.2.5 Implementation

```
// Beat counter and address tracking  
logic [7:0] r_beat_count;  
logic [7:0] r_arlen_saved;  
logic [ADDR_WIDTH-1:0] r_current_addr;  
  
// State machine  
always_ff @(posedge clk or negedge rst_n) begin  
    if (!rst_n) begin  
        r_state <= IDLE;  
        r_beat_count <= '0;  
    end else begin  
        case (r_state)  
            IDLE: begin  
                if (s_arvalid && s_arready) begin
```

```

        r_arlen_saved <= s_arlen;
        r_current_addr <= s_araddr;
        if (s_arlen == 0) begin
            // Single beat - passthrough
            r_state <= WAIT_R;
        end else begin
            r_state <= DECOMPOSE;
            r_beat_count <= 8'd1;
        end
    end
end

DECOMPOSE: begin
    if (m_arvalid && m_arready) begin
        r_current_addr <= r_current_addr + (1 <<
s_arsize);
        r_beat_count <= r_beat_count + 1;
        if (r_beat_count == r_arlen_saved) begin
            r_state <= WAIT_R;
        end
    end
end

WAIT_R: begin
    if (s_rvalid && s_rready && s_rlast) begin
        r_state <= IDLE;
    end
end
endcase
end
end

// AXIL4 AR generation
assign m_arvalid = (r_state == DECOMPOSE) || (r_state == IDLE &&
s_arvalid);
assign m_araddr = r_current_addr;

// R aggregation
assign s_rlast = (r_beat_count == r_arlen_saved) || (r_arlen_saved ==
0);

```

11.3 3.2.3 Write Path (axi4_to_axil4_wr)

11.3.1 Block Diagram

11.3.2 Figure 3.3: AXI4 to AXI4-Lite Write Path

AXI4 to AXIL4 Write

AXI4 to AXIL4 Write

11.3.3 Operation

Single-Beat (AWLEN=0):

Cycle 0: AW+W accepted (passthrough)
Cycle 1: B returned (passthrough)
Total: 0 extra cycles

Multi-Beat (AWLEN=N-1):

Cycle 0: AW[0] + W[0] issued
Cycle 1: B[0] received, AW[1] + W[1] issued
...
Cycle 2N-1: B[N-1] received
Total: 2N cycles

11.3.4 AW/W Synchronization Challenge

AXI4 allows AW and W to arrive in any order: - AW before W - W before AW - Interleaved

11.3.5 Solution: Dual Accept Logic

// Track AW and W arrival independently

```
logic r_aw_accepted;  
logic r_w_accepted;
```

// Accept both when ready to issue

```
always_ff @(posedge clk) begin  
    if (w_issue_axil4) begin  
        r_aw_accepted <= 1'b0;  
        r_w_accepted <= 1'b0;  
    end else begin  
        if (s_awvalid && s_awready)  
            r_aw_accepted <= 1'b1;  
        if (s_wvalid && s_wready)  
            r_w_accepted <= 1'b1;  
    end  
end
```

// Issue AXIL4 when both available

```

assign w_issue_axi4 = (r_aw_accepted || s_awvalid) &&
                        (r_w_accepted || s_wvalid) &&
                        m_ready;

```

11.3.6 Response Aggregation

```

// Track worst response in burst

```

```

logic [1:0] r_worst_bresp;

```

```

always_ff @(posedge clk) begin

```

```

    if (r_state == IDLE)

```

```

        r_worst_bresp <= 2'b00; // OKAY

```

```

    else if (m_bvalid && m_bready)

```

```

        r_worst_bresp <= (m_bresp > r_worst_bresp) ? m_bresp :

```

```

        r_worst_bresp;

```

```

end

```

```

// Return worst response on final beat

```

```

assign s_bresp = (r_beat_count == r_awlen_saved) ?

```

```

    ((m_bresp > r_worst_bresp) ? m_bresp : r_worst_bresp)

```

```

:

```

```

    r_worst_bresp;

```

11.4 3.2.4 Bidirectional Wrapper (axi4_to_axil4)

11.4.1 Composition Pattern

```
module axi4_to_axil4 #(
    parameter int DATA_WIDTH = 32,
    parameter int ADDR_WIDTH = 32,
    parameter int ID_WIDTH    = 4
) (
    // ... ports
);

    // Instantiate read path
    axi4_to_axil4_rd #(
        .DATA_WIDTH(DATA_WIDTH),
        .ADDR_WIDTH(ADDR_WIDTH),
        .ID_WIDTH(ID_WIDTH)
    ) u_rd (
        // ... read channel connections
    );

    // Instantiate write path
    axi4_to_axil4_wr #(
        .DATA_WIDTH(DATA_WIDTH),
        .ADDR_WIDTH(ADDR_WIDTH),
        .ID_WIDTH(ID_WIDTH)
    ) u_wr (
        // ... write channel connections
    );

endmodule
```

11.5 3.2.5 Resource Utilization

Table 3.6: AXI4 to AXIL4 Resources

Module	Registers	LUTs	BRAM
axi4_to_axil4_r d	~120	~180	0
axi4_to_axil4_ wr	~150	~220	0
axi4_to_axil4 (combined)	~270	~400	0

11.6 3.2.6 Performance Analysis

11.6.1 Throughput

Table 3.7: AXI4 to AXIL4 Throughput

Transaction Type	Throughput
Single-beat	100% (passthrough)
2-beat burst	50%
4-beat burst	50%
N-beat burst	~50%

11.6.2 Latency

Table 3.8: AXI4 to AXIL4 Latency

Transaction Type	Latency
Single-beat	0 extra cycles
N-beat burst	2N - 1 cycles

11.7 3.2.7 Test Coverage

Test Suite: 42 tests passing

Table 3.9: Test Coverage Summary

Test Category	Tests	Status
Single-beat read	4	Pass
Multi-beat read	6	Pass
Single-beat write	4	Pass
Multi-beat write	6	Pass
Mixed traffic	8	Pass
Error injection	6	Pass
Edge cases	8	Pass

Next: [AXI4-Lite to AXI4](#)

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12 3.3 AXI4-Lite to AXI4 Converter

The **axil4_to_axi4** converter family upgrades AXI4-Lite single-beat transactions to full AXI4 protocol by adding default burst signals.

12.1 3.3.1 Module Organization

```
axil4_to_axi4.sv      # Full bidirectional wrapper
├─ axil4_to_axi4_rd.sv # Read path converter
└─ axil4_to_axi4_wr.sv # Write path converter
```

12.2 3.3.2 Design Philosophy

Zero-Overhead Upgrade: - Purely combinational logic - No state machines or buffers - Adds default values for missing AXI4 signals

Why This Works: - AXI4-Lite is a subset of AXI4 - All AXI4-Lite transactions are single-beat - Missing AXI4 signals have well-defined defaults

12.3 3.3.3 Signal Mapping

12.3.1 Address Channel Signals

Table 3.10: AR Channel Mapping

AXI4-Lite Signal	AXI4 Signal	Default/Mapping
ARADDR	ARADDR	Passthrough
ARPROT	ARPROT	Passthrough
ARVALID	ARVALID	Passthrough
ARREADY	ARREADY	Passthrough
-	ARLEN	8'h00 (single beat)
-	ARSIZE	$\$clog2(DATA_WIDTH/8)$
-	ARBURST	2'b01 (INCR)
-	ARLOCK	1'b0 (normal)
-	ARCACHE	4'b0000 (non-cacheable)
-	ARQOS	4'b0000 (no QoS)
-	ARID	Configurable default

12.3.2 Data Channel Signals

Table 3.11: R Channel Mapping

AXI4-Lite Signal	AXI4 Signal	Default/Mapping
RDATA	RDATA	Passthrough
RRESP	RRESP	Passthrough
RVALID	RVALID	Passthrough
RREADY	RREADY	Passthrough
-	RLAST	1'b1 (always last)
-	RID	Matches ARID

12.4 3.3.4 Read Path (axil4_to_axi4_rd)

12.4.1 Block Diagram

12.4.2 Figure 3.4: AXI4-Lite to AXI4 Read Path

AXI4 to AXI4 Read

AXI4 to AXI4 Read

12.4.3 Implementation

```
module axil4_to_axi4_rd #(
    parameter int DATA_WIDTH = 32,
    parameter int ADDR_WIDTH = 32,
    parameter int ID_WIDTH = 4,
    parameter logic [ID_WIDTH-1:0] DEFAULT_ARID = '0
) (
    // AXI4-Lite slave interface (input)
    input logic s_arvalid,
    output logic s_arready,
    input logic [ADDR_WIDTH-1:0] s_araddr,
    input logic [2:0] s_arprot,

    output logic s_rvalid,
    input logic s_rready,
    output logic [DATA_WIDTH-1:0] s_rdata,
    output logic [1:0] s_rresp,

    // AXI4 master interface (output)
    output logic m_arvalid,
    input logic m_arready,
    output logic [ADDR_WIDTH-1:0] m_araddr,
    output logic [7:0] m_arlen,
    output logic [2:0] m_arsize,
    output logic [1:0] m_arburst,
    output logic m_arlock,
    output logic [3:0] m_arcache,
    output logic [2:0] m_arprot,
    output logic [3:0] m_arqos,
    output logic [ID_WIDTH-1:0] m_arid,

    input logic m_rvalid,
    output logic m_rready,
    input logic [DATA_WIDTH-1:0] m_rdata,
    input logic [1:0] m_rresp,
    input logic m_rlast,
```

```

);    input  logic [ID_WIDTH-1:0]    m_rid

// AR channel - passthrough with defaults
assign m_arvalid = s_arvalid;
assign s_arready = m_arready;
assign m_araddr  = s_araddr;
assign m_arprot  = s_arprot;

// AXI4 burst defaults
assign m_arlen   = 8'h00;           // Single beat
assign m_arsize  = $clog2(DATA_WIDTH/8); // Full width
assign m_arburst = 2'b01;           // INCR
assign m_arlock  = 1'b0;           // Normal access
assign m_arcache = 4'b0000;         // Non-cacheable
assign m_arqos   = 4'b0000;         // No QoS
assign m_arid    = DEFAULT_ARID;    // Configurable ID

// R channel - passthrough (ignore RLAST and RID)
assign s_rvalid  = m_rvalid;
assign m_rready  = s_rready;
assign s_rdata   = m_rdata;
assign s_rresp   = m_rresp;

```

endmodule

Key Points: - No registers - purely combinational - RLAST from AXI4 is ignored (always 1 for AXIL4) - RID from AXI4 is ignored (no ID tracking in AXIL4)

12.5 3.3.5 Write Path (axil4_to_axi4_wr)

12.5.1 Block Diagram

12.5.2 Figure 3.5: AXI4-Lite to AXI4 Write Path

AXIL4 to AXI4 Write

AXIL4 to AXI4 Write

12.5.3 Implementation

```
module axil4_to_axi4_wr #(
    parameter int DATA_WIDTH = 32,
    parameter int ADDR_WIDTH = 32,
    parameter int ID_WIDTH = 4,
    parameter logic [ID_WIDTH-1:0] DEFAULT_AWID = '0
) (
    // AXI4-Lite slave interface (input)
    input logic s_awvalid,
    output logic s_awready,
    input logic [ADDR_WIDTH-1:0] s_awaddr,
    input logic [2:0] s_awprot,

    input logic s_wvalid,
    output logic s_wready,
    input logic [DATA_WIDTH-1:0] s_wdata,
    input logic [DATA_WIDTH/8-1:0] s_wstrb,

    output logic s_bvalid,
    input logic s_bready,
    output logic [1:0] s_bresp,

    // AXI4 master interface (output)
    output logic m_awvalid,
    input logic m_awready,
    output logic [ADDR_WIDTH-1:0] m_awaddr,
    output logic [7:0] m_awlen,
    output logic [2:0] m_awsz,
    output logic [1:0] m_awburst,
    output logic m_awlock,
    output logic [3:0] m_awcache,
    output logic [2:0] m_awprot,
    output logic [3:0] m_awqos,
    output logic [ID_WIDTH-1:0] m_awid,

    output logic m_wvalid,
    input logic m_wready,
```

```

    output logic [DATA_WIDTH-1:0] m_wdata,
    output logic [DATA_WIDTH/8-1:0] m_wstrb,
    output logic m_wlast,

    input logic m_bvalid,
    output logic m_bready,
    input logic [1:0] m_bresp,
    input logic [ID_WIDTH-1:0] m_bid
);

// AW channel
assign m_awvalid = s_awvalid;
assign s_awready = m_awready;
assign m_awaddr = s_awaddr;
assign m_awprot = s_awprot;
assign m_awlen = 8'h00;
assign m_awsz = $clog2(DATA_WIDTH/8);
assign m_awburst = 2'b01;
assign m_awlock = 1'b0;
assign m_awcache = 4'b0000;
assign m_awqos = 4'b0000;
assign m_awid = DEFAULT_AWID;

// W channel
assign m_wvalid = s_wvalid;
assign s_wready = m_wready;
assign m_wdata = s_wdata;
assign m_wstrb = s_wstrb;
assign m_wlast = 1'b1; // Always last (single beat)

// B channel
assign s_bvalid = m_bvalid;
assign m_bready = s_bready;
assign s_bresp = m_bresp;

endmodule

```

12.6 3.3.6 Bidirectional Wrapper

```
module axil4_to_axi4 #(
    parameter int DATA_WIDTH = 32,
    parameter int ADDR_WIDTH = 32,
    parameter int ID_WIDTH = 4,
    parameter logic [ID_WIDTH-1:0] DEFAULT_ID = '0
) (
    // ... all port declarations
);

axil4_to_axi4_rd #(
    .DATA_WIDTH(DATA_WIDTH),
    .ADDR_WIDTH(ADDR_WIDTH),
    .ID_WIDTH(ID_WIDTH),
    .DEFAULT_ARID(DEFAULT_ID)
) u_rd (/* connections */);

axil4_to_axi4_wr #(
    .DATA_WIDTH(DATA_WIDTH),
    .ADDR_WIDTH(ADDR_WIDTH),
    .ID_WIDTH(ID_WIDTH),
    .DEFAULT_AWID(DEFAULT_ID)
) u_wr (/* connections */);

endmodule
```

12.7 3.3.7 Resource Utilization

Table 3.12: AXIL4 to AXI4 Resources

Module	Registers	LUTs
axil4_to_axi4_rd	0	~50
axil4_to_axi4_wr	0	~60
axil4_to_axi4 (combined)	0	~110

Note: Zero registers - purely combinational logic.

12.8 3.3.8 Performance

Table 3.13: AXIL4 to AXI4 Performance

Metric	Value
Latency	0 cycles
Throughput	100%
Max frequency	Wire speed

12.9 3.3.9 Test Coverage

Test Suite: 14 tests passing

Table 3.14: Test Coverage Summary

Test Category	Tests	Status
Single-beat read	3	Pass
Single-beat write	3	Pass
Mixed traffic	4	Pass
Default ID verification	2	Pass
Edge cases	2	Pass

12.10 3.3.10 Usage Example

```
// Upgrade simple register block to AXI4 fabric
axil4_to_axi4 #(
    .DATA_WIDTH(32),
    .ADDR_WIDTH(32),
    .ID_WIDTH(4),
    .DEFAULT_ID(4'h5) // Unique ID for this IP
) u_protocol_upgrade (
    // Connect AXIL4 register block
    .s_arvalid (reg_block_arvalid),
    .s_arready (reg_block_arready),
    .s_araddr  (reg_block_araddr),
    // ... other AXIL4 signals

    // Connect to AXI4 crossbar
    .m_arvalid (xbar_arvalid),
    .m_arready (xbar_arready),
    .m_araddr  (xbar_araddr),
    .m_arlen   (xbar_arlen),
    // ... other AXI4 signals
);
```

Next: [AXI4 to APB](#)

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13 3.4 AXI4 to APB Converter

The **axi4_to_apb4_convert** module provides full protocol translation from AXI4 to APB, enabling AXI4 masters to access APB peripherals.

13.1 3.4.1 Purpose

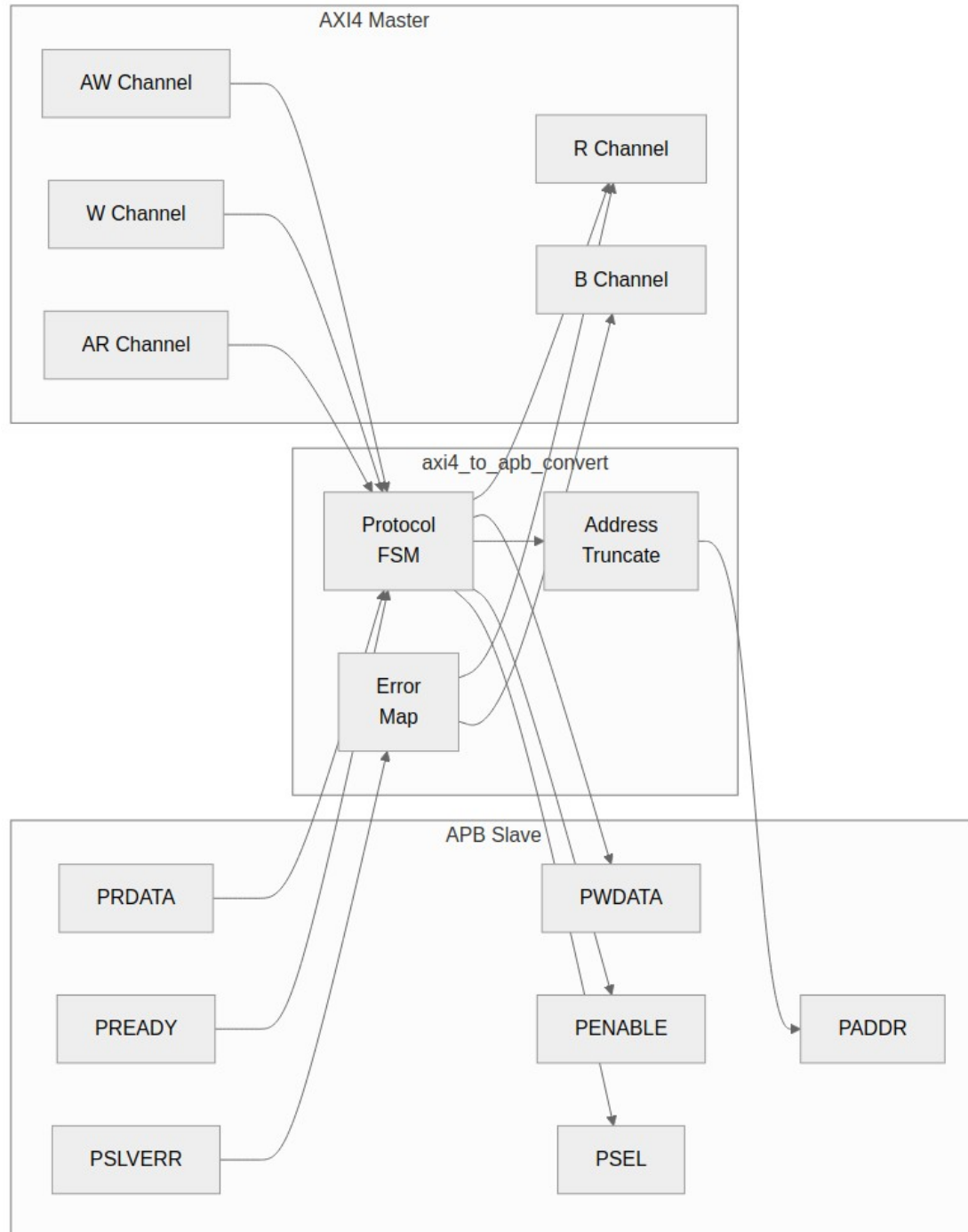
AXI4 and APB differ on just about every axis:

Table 3.15: AXI4 vs APB Comparison

Aspect	AXI4	APB
Channels	5 (AW, W, B, AR, R)	1 (combined)
Phases	Pipelined	2-phase (setup, access)
Bursts	Up to 256 beats	Single transfer
Address width	Up to 64 bits	Typically 32 bits
Data width	8-1024 bits	8-32 bits

13.2 3.4.2 Block Diagram

13.2.1 Figure 3.6: AXI4 to APB Converter



AXI4 to APB

13.3 3.4.3 Interface Specification

13.3.1 Parameters

Table 3.16: AXI4 to APB Parameters

Parameter	Type	Default	Description
AXI_ADDR_WI DTH	int	64	AXI4 address width
AXI_DATA_WI DTH	int	32	AXI4 data width
AXI_ID_WIDTH	int	4	AXI4 ID width
APB_ADDR_WI DTH	int	32	APB address width
APB_DATA_WI DTH	int	32	APB data width

13.3.2 Ports

```
module axi4_to_apb4_convert #(
    parameter int AXI_ADDR_WIDTH = 64,
    parameter int AXI_DATA_WIDTH = 32,
    parameter int AXI_ID_WIDTH   = 4,
    parameter int APB_ADDR_WIDTH = 32,
    parameter int APB_DATA_WIDTH = 32
) (
    input  logic clk,
    input  logic rst_n,

    // AXI4 slave interface
    // AW channel
    input  logic          s_awvalid,
    output logic          s_awready,
    input  logic [AXI_ADDR_WIDTH-1:0] s_awaddr,
    input  logic [7:0] s_awlen,
    input  logic [AXI_ID_WIDTH-1:0] s_awid,

    // W channel
    input  logic          s_wvalid,
    output logic          s_wready,
    input  logic [AXI_DATA_WIDTH-1:0] s_wdata,
    input  logic [AXI_DATA_WIDTH/8-1:0] s_wstrb,
    input  logic          s_wlast,
```

```

// B channel
output logic                                s_bvalid,
input  logic                                s_bready,
output logic [AXI_ID_WIDTH-1:0]            s_bid,
output logic [1:0]                          s_bresp,

// AR channel
input  logic                                s_arvalid,
output logic                                s_arready,
input  logic [AXI_ADDR_WIDTH-1:0]          s_araddr,
input  logic [7:0]                          s_arlen,
input  logic [AXI_ID_WIDTH-1:0]            s_arid,

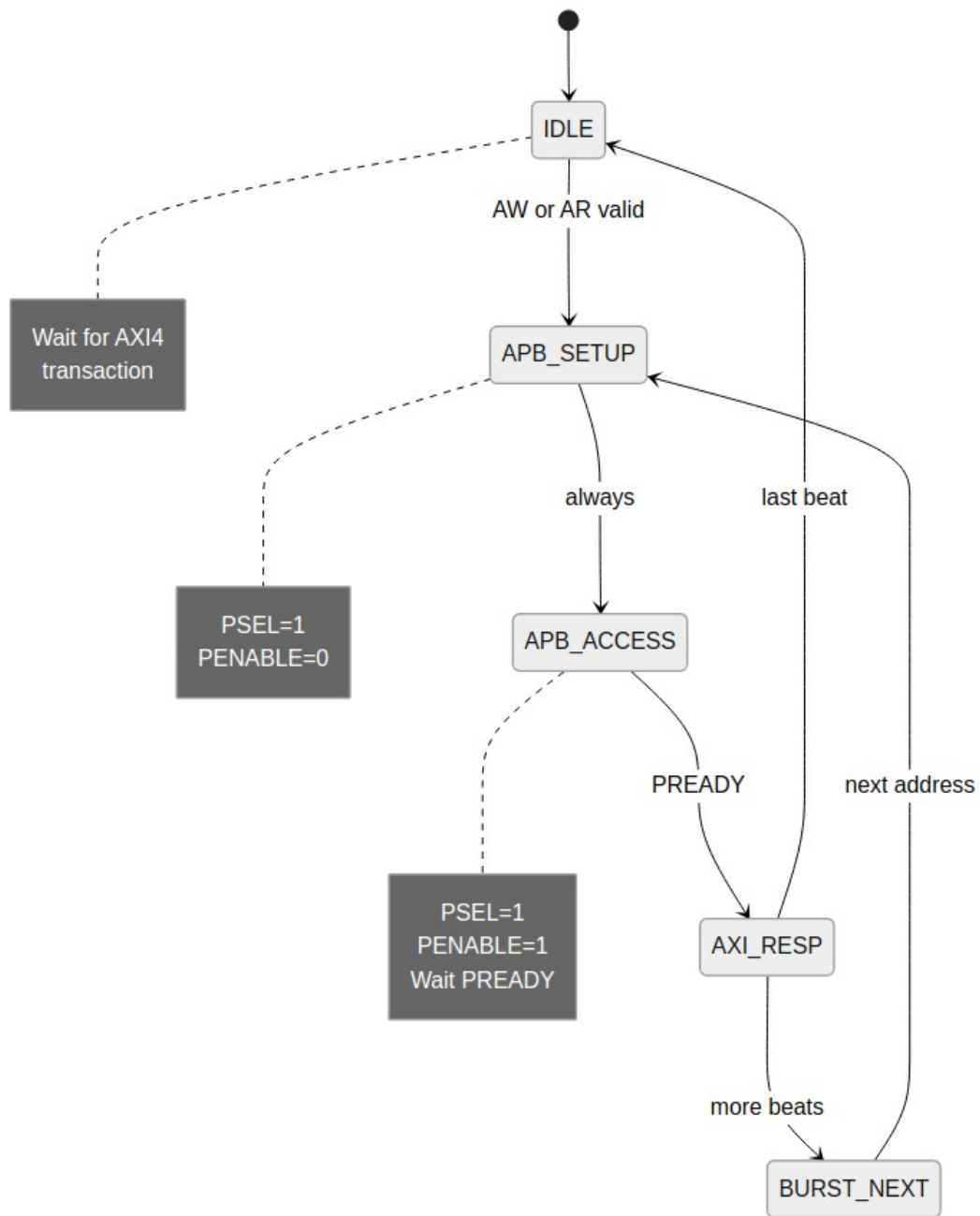
// R channel
output logic                                s_rvalid,
input  logic                                s_rready,
output logic [AXI_DATA_WIDTH-1:0]          s_rdata,
output logic [AXI_ID_WIDTH-1:0]            s_rid,
output logic [1:0]                          s_rresp,
output logic                                s_rlast,

// APB master interface
output logic                                psel,
output logic                                penable,
output logic                                pwrite,
output logic [APB_ADDR_WIDTH-1:0]          paddr,
output logic [APB_DATA_WIDTH-1:0]          pwrdata,
output logic [APB_DATA_WIDTH/8-1:0]        pstrb,
input  logic                                pready,
input  logic [APB_DATA_WIDTH-1:0]          prdata,
input  logic                                pslverr
);

```

13.4 3.4.4 State Machine

13.4.1 Figure 3.7: AXI4 to APB FSM



AXI4 to APB FSM

13.4.2 States

```
typedef enum logic [2:0] {  
    IDLE          = 3'b000, // Wait for AXI4 transaction  
    APB_SETUP     = 3'b001, // APB setup phase  
    APB_ACCESS    = 3'b010, // APB access phase (wait PREADY)  
    AXI_RESP_B    = 3'b011, // Send AXI4 B response  
    AXI_RESP_R    = 3'b100, // Send AXI4 R response  
    BURST_NEXT    = 3'b101  // Next beat in burst  
} apb_state_t;
```

13.4.3 Transitions

Table 3.17: FSM Transitions

Current State	Condition	Next State
IDLE	s_awvalid	APB_SETUP (write)
IDLE	s_arvalid	APB_SETUP (read)
APB_SETUP	always	APB_ACCESS
APB_ACCESS	pready && is_write	AXI_RESP_B
APB_ACCESS	pready && is_read	AXI_RESP_R
AXI_RESP_B	s_bready && ! more_beats	IDLE
AXI_RESP_B	s_bready && more_beats	BURST_NEXT
AXI_RESP_R	s_rready && ! more_beats	IDLE
AXI_RESP_R	s_rready && more_beats	BURST_NEXT
BURST_NEXT	always	APB_SETUP

13.5 3.4.5 Burst Handling

13.5.1 Burst Decomposition

AXI4 bursts are decomposed into sequential APB transfers:

AXI4: AWADDR=0x1000, AWLEN=3 (4 beats)

APB sequence:

Transfer 0: PADDR=0x1000

Transfer 1: PADDR=0x1004

Transfer 2: PADDR=0x1008

Transfer 3: PADDR=0x100C

13.5.2 Address Calculation

```
// Calculate next address for INCR burst
logic [APB_ADDR_WIDTH-1:0] r_current_addr;
logic [2:0] r_awsz;

always_ff @(posedge clk) begin
    if (r_state == APB_ACCESS && pready) begin
        r_current_addr <= r_current_addr + (1 << r_awsz);
    end
end
```

13.6 3.4.6 Address Width Adaptation

13.6.1 64-bit to 32-bit Conversion

// Truncate upper address bits

assign paddr = s_awaddr[APB_ADDR_WIDTH-1:0];

// Optional: Check for out-of-range access

wire w_addr_oor = |s_awaddr[AXI_ADDR_WIDTH-1:APB_ADDR_WIDTH];

13.7 3.4.7 Error Response Mapping

Table 3.18: Error Mapping

APB Signal	AXI4 Response
PSLVERR = 0	OKAY (2'b00)
PSLVERR = 1	SLVERR (2'b10)

13.7.1 Error Aggregation

// Track worst error in burst

```
logic r_error_seen;
```

```
always_ff @(posedge clk) begin
```

```
    if (r_state == IDLE)
```

```
        r_error_seen <= 1'b0;
```

```
    else if (r_state == APB_ACCESS && pready && pslverr)
```

```
        r_error_seen <= 1'b1;
```

```
end
```

// Final response

```
assign s_bresp = r_error_seen ? 2'b10 : 2'b00;
```

13.8 3.4.8 Implementation

13.8.1 Core FSM

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_state <= IDLE;
        r_is_write <= 1'b0;
    end else begin
        case (r_state)
            IDLE: begin
                if (s_awvalid && s_wvalid) begin
                    r_state <= APB_SETUP;
                    r_is_write <= 1'b1;
                    r_current_addr <= s_awaddr[APB_ADDR_WIDTH-1:0];
                    r_beat_count <= '0;
                    r_awlen <= s_awlen;
                    r_awid <= s_awid;
                end else if (s_arvalid) begin
                    r_state <= APB_SETUP;
                    r_is_write <= 1'b0;
                    r_current_addr <= s_araddr[APB_ADDR_WIDTH-1:0];
                    r_beat_count <= '0;
                    r_arlen <= s_arlen;
                    r_arid <= s_arid;
                end
            end
        end

        APB_SETUP: begin
            r_state <= APB_ACCESS;
        end

        APB_ACCESS: begin
            if (pready) begin
                r_rdata_saved <= prdata;
                r_error_seen <= r_error_seen || pslverr;
                if (r_is_write)
                    r_state <= AXI_RESP_B;
                else
                    r_state <= AXI_RESP_R;
            end
        end

        AXI_RESP_B: begin
            if (s_bready) begin
                if (r_beat_count == r_awlen)
                    r_state <= IDLE;
                else

```

```

        r_state <= BURST_NEXT;
    end
end

AXI_RESP_R: begin
    if (s_rready) begin
        if (r_beat_count == r_arlen)
            r_state <= IDLE;
        else
            r_state <= BURST_NEXT;
        end
    end
end

BURST_NEXT: begin
    r_beat_count <= r_beat_count + 1;
    r_current_addr <= r_current_addr + (1 << r_size);
    r_state <= APB_SETUP;
end
endcase
end
end
end

```

13.9 3.4.9 Resource Utilization

State machine:	~50 LUTs, ~20 regs
Address logic:	~30 LUTs, ~40 regs
Data buffering:	~10 LUTs, ~70 regs
Control:	~60 LUTs, ~20 regs

Total: ~150 LUTs, ~150 regs

13.10 3.4.10 Performance

13.10.1 Timing Analysis

Table 3.19: APB Converter Timing

Operation	Cycles
Single write	3-4 (setup + access + B)
Single read	3-4 (setup + access + R)
N-beat write burst	$3N + 1$
N-beat read burst	$3N + 1$

13.10.2 Throughput

Best case: 1 transfer per 3 cycles **With slow PREADY:** Additional cycles per transfer

3.4.11 Usage Example

```
axi4_to_apb4_convert #(
    .AXI_ADDR_WIDTH(64),
    .AXI_DATA_WIDTH(32),
    .AXI_ID_WIDTH(4),
    .APB_ADDR_WIDTH(32),
    .APB_DATA_WIDTH(32)
) u_axi2apb4 (
    .clk      (aclk),
    .rst_n    (aresetn),

    // AXI4 slave (from CPU)
    .s_awvalid (cpu_awvalid),
    .s_awready (cpu_awready),
    // ... other AXI4 signals

    // APB master (to peripherals)
    .psel      (uart_psel),
    .penable   (uart_penable),
    .pwrite    (uart_pwrite),
    .paddr     (uart_paddr),
    .pwrdata   (uart_pwdata),
    .pready    (uart_pready),
    .prdata    (uart_prdata),
    .pslverr   (uart_pslverr)
);
```

Next: PeakRDL Adapter

14 AXI4 to APB5 Shim

Module: axi4_to_apb5_shim.sv **Filelist:** rtl/filelists/axi4_to_apb5_shim.f (-f's the APB4 shim's closure)

14.1 Design

APB5 keeps the APB4 transfer protocol — PSEL/PENABLE phases, PREADY, PSLVERR are unchanged — and adds only sideband: requester-driven user signals and completer-driven wakeup/user signals. So this block is a thin **pin-superset wrapper** over `axi4_to_apb4_shim`: the entire protocol engine (CDC, cmd/rsp FIFOs, burst decomposition, response assembly) is the APB4 shim instantiated unchanged, with all 58 ports and 16 parameters forwarded 1:1.

The APB5 additions on the requester surface:

Signal	Dir	Handling
m_apb_PAUSER[APB_AUSER_WIDTH-1:0]	out	tied '0 — nothing upstream sources it (AXI USER bits do not map onto APB user semantics)
m_apb_PWUSER[APB_WUSER_WIDTH-1:0]	out	tied '0
m_apb_PWAKEUP	in	accepted and terminated
m_apb_PRUSER[APB_RUSER_WIDTH-1:0]	in	accepted and terminated
m_apb_PBUSER[APB_BUSER_WIDTH-1:0]	in	accepted and terminated

User-signal widths default to 1 and match `rtl/amba/apb5/apb5_slave.sv`, which this requester drops onto pin-for-pin (including the repo's convention that PWAKEUP rides completer→requester).

14.2 Rationale

Deriving the wrapper from the APB4 shim's actual port surface (rather than reimplementing the conversion) means the APB4 engine's fixes and its dependency closure are inherited automatically — the closure filelist simply -f's the APB4 shim's. When a future consumer needs real PAUSER/PWUSER sourcing or PWAKEUP-gated clocking, those grow here without touching the conversion core.

14.3 Consumers

The bridge generator instantiates this shim for `protocol = "apb5"` slaves (BRIDGE-002 A5-3c) through the same component path as the APB4 shim — the `Axi4ToApbShim` component takes `protocol='apb5'` and wires the five extra pairs. Verified by `projects/components/bridge/dv/tests/test_bridge_1x2_rw_apb5.py` (APB4 BFM legally drives the port: same transfer protocol).

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15 3.5 PeakRDL Adapter

The **peakrdl_to_cmdrsp** module adapts PeakRDL-generated register interfaces to a custom command/response protocol, enabling protocol decoupling and flexible register implementations.

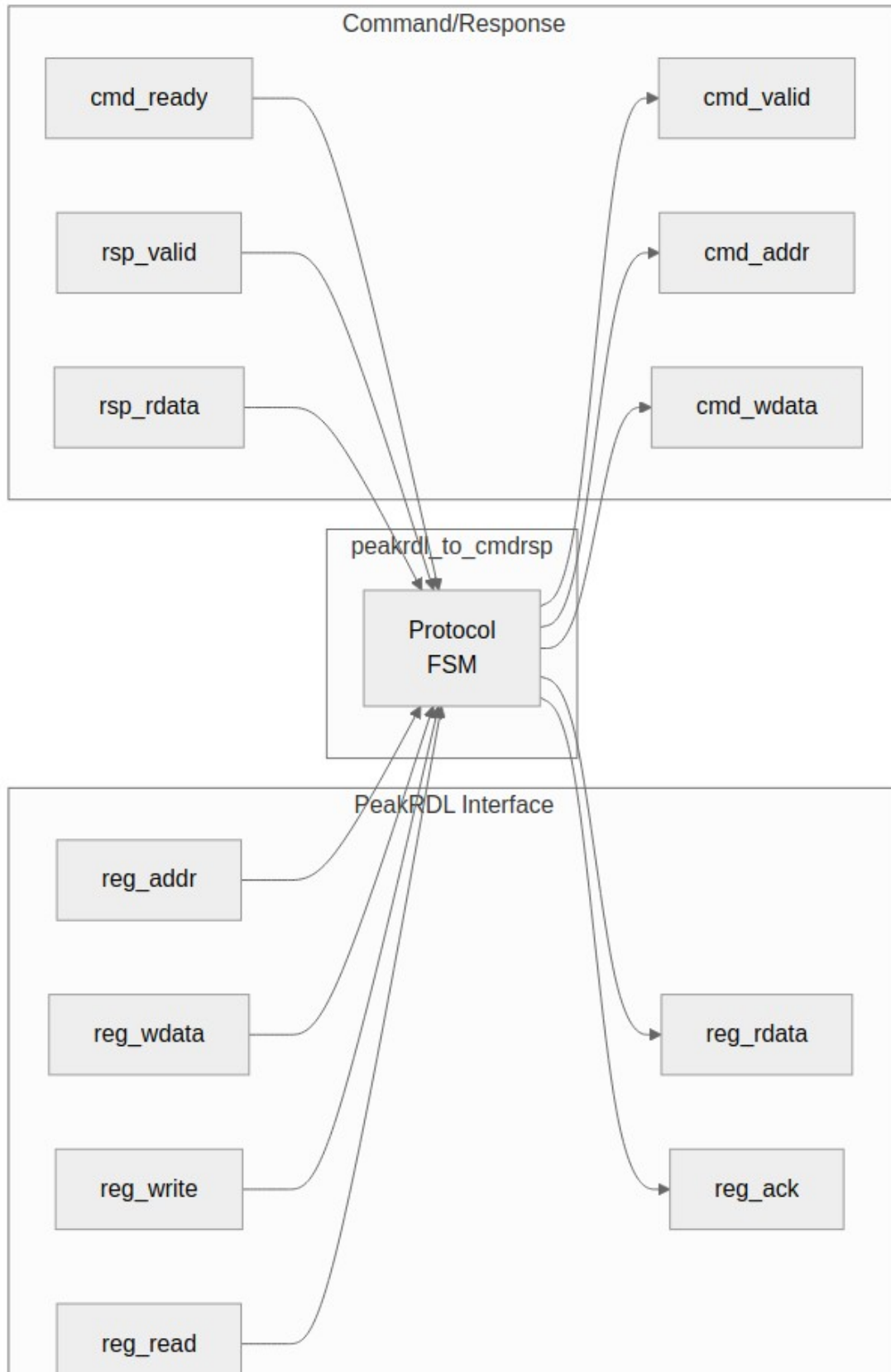
15.1 3.5.1 Purpose

PeakRDL generates register blocks with an APB-style interface. This adapter sits behind that interface and:

1. Decouples the register interface from the implementation
2. Provides a clean handshake protocol
3. Enables pipelined register access
4. Supports custom control logic integration

15.2 3.5.2 Block Diagram

15.2.1 Figure 3.8: PeakRDL Adapter



PeakRDL Adapter

15.3 3.5.3 Interface Specification

15.3.1 Parameters

Table 3.20: PeakRDL Adapter Parameters

Parameter	Type	Default	Description
ADDR_WIDTH	int	32	Address width
DATA_WIDTH	int	32	Data width

15.3.2 Ports

```
module peakrdl_to_cmdrsp #(
    parameter int ADDR_WIDTH = 32,
    parameter int DATA_WIDTH = 32
) (
    input logic clk,
    input logic rst_n,

    // Register interface (from PeakRDL)
    input logic [ADDR_WIDTH-1:0] reg_addr,
    input logic [DATA_WIDTH-1:0] reg_wdata,
    input logic reg_write,
    input logic reg_read,
    output logic [DATA_WIDTH-1:0] reg_rdata,
    output logic reg_error,
    output logic reg_ack,

    // Command interface (output)
    output logic cmd_valid,
    input logic cmd_ready,
    output logic [ADDR_WIDTH-1:0] cmd_addr,
    output logic [DATA_WIDTH-1:0] cmd_wdata,
    output logic cmd_write,

    // Response interface (input)
    input logic rsp_valid,
    output logic rsp_ready,
    input logic [DATA_WIDTH-1:0] rsp_rdata,
    input logic rsp_error
);
```

15.4 3.5.4 Operation

15.4.1 Write Transaction

Cycle 0: reg_write asserted
 cmd_valid = 1, cmd_write = 1
Cycle 1: cmd_ready = 1 (downstream accepts)
 Wait for response
Cycle N: rsp_valid = 1
 reg_ack = 1
Cycle N+1: Transaction complete

15.4.2 Read Transaction

Cycle 0: reg_read asserted
 cmd_valid = 1, cmd_write = 0
Cycle 1: cmd_ready = 1 (downstream accepts)
 Wait for response
Cycle N: rsp_valid = 1
 reg_rdata = rsp_rdata
 reg_ack = 1
Cycle N+1: Transaction complete

15.5 3.5.5 Implementation

```
// State machine
typedef enum logic [1:0] {
    IDLE    = 2'b00,
    CMD     = 2'b01,
    RSP     = 2'b10
} state_t;

state_t r_state;
logic r_is_write;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_state <= IDLE;
    end else begin
        case (r_state)
            IDLE: begin
                if (reg_write || reg_read) begin
                    r_state <= CMD;
                    r_is_write <= reg_write;
                end
            end

            CMD: begin
                if (cmd_ready) begin
                    r_state <= RSP;
                end
            end

            RSP: begin
                if (rsp_valid) begin
                    r_state <= IDLE;
                end
            end
        endcase
    end
end

// Command interface
assign cmd_valid = (r_state == CMD);
assign cmd_addr = reg_addr;
assign cmd_wdata = reg_wdata;
assign cmd_write = r_is_write;

// Response interface
assign rsp_ready = (r_state == RSP);
```

```
// Register interface  
assign reg_rdata = rsp_rdata;  
assign reg_error = rsp_error;  
assign reg_ack = (r_state == RSP) && rsp_valid;
```

15.6 3.5.6 Resource Utilization

State machine: ~20 LUTs, ~10 regs
Data paths: ~10 LUTs, ~40 regs
Control: ~20 LUTs, ~5 regs

Total: ~50 LUTs, ~55 regs

15.7 3.5.7 Use Cases

15.7.11. PeakRDL to Custom Control

PeakRDL Registers → Adapter → Custom State Machine
→ Hardware Accelerator
→ Debug Controller

15.7.22. Register Access Logging

PeakRDL Registers → Adapter → Logger → Registers
↓
Log Buffer

15.7.33. Pipeline Insertion

PeakRDL Registers → Adapter → Pipeline → Slow Registers
(for timing)

15.8 3.5.8 Integration Example

```
// Instantiate PeakRDL-generated register block
```

```
my_regs u_regs (  
    .clk      (clk),  
    .rst_n    (rst_n),
```

```
// APB-style interface from CPU
```

```
.s_apb_psel    (apb_psel),  
.s_apb_penable (apb_penable),  
// ... other APB signals
```

```
// Register interface to adapter
```

```
.reg_addr    (reg_addr),  
.reg_wdata    (reg_wdata),  
.reg_write    (reg_write),  
.reg_read     (reg_read),  
.reg_rdata    (reg_rdata),  
.reg_error    (reg_error),  
.reg_ack      (reg_ack)
```

```
);
```

```
// Adapter to custom protocol
```

```
peakrdl_to_cmdrsp #(  
    .ADDR_WIDTH(32),
```

```
.DATA_WIDTH(32)
```

```
) u_adapter (  
    .clk      (clk),
```

```
.rst_n    (rst_n),
```

```
// From PeakRDL registers
```

```
.reg_addr    (reg_addr),  
.reg_wdata    (reg_wdata),  
.reg_write    (reg_write),  
.reg_read     (reg_read),  
.reg_rdata    (reg_rdata),  
.reg_error    (reg_error),  
.reg_ack      (reg_ack),
```

```
// To custom control logic
```

```
.cmd_valid    (ctrl_cmd_valid),  
.cmd_ready    (ctrl_cmd_ready),  
.cmd_addr     (ctrl_cmd_addr),  
.cmd_wdata    (ctrl_cmd_wdata),  
.cmd_write    (ctrl_cmd_write),
```

```
.rsp_valid    (ctrl_rsp_valid),
```

```
        .rsp_ready    (ctrl_rsp_ready),  
        .rsp_rdata    (ctrl_rsp_rdata),  
        .rsp_error    (ctrl_rsp_error)  
    );
```

Next: [Chapter 4: FSM Design](#)

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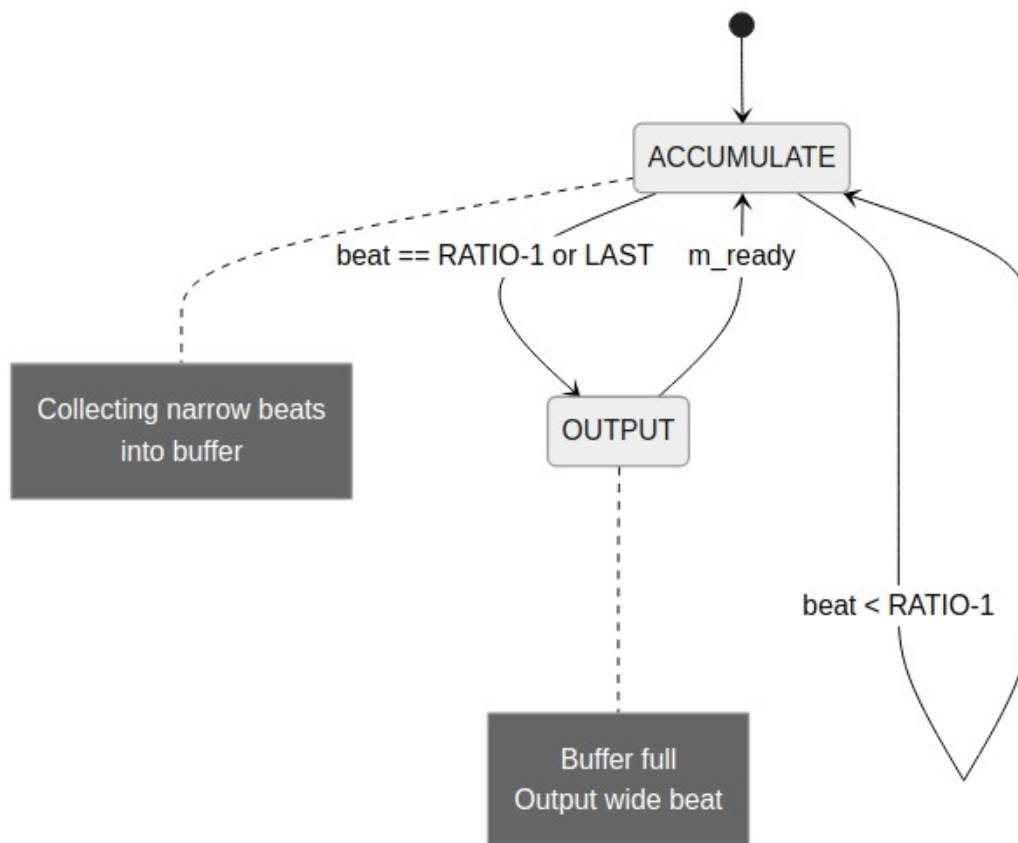
16 4.1 Width Converter FSMs

These are the state machines inside the width converters.

16.1 4.1.1 Upsize FSM

The **axi_data_upsize** module uses a simple accumulation state machine.

16.1.1 Figure 4.1: Upsize FSM



Upsize FSM

16.1.2 States

Table 4.1: Upsize FSM States

State	Description
ACCUMULATE	Collecting narrow beats into buffer
OUTPUT	Buffer full, outputting wide beat

16.1.3 Transitions

ACCUMULATE:

- s_valid && count < RATIO-1 → stay, increment count
- s_valid && (count == RATIO-1 || s_last) → OUTPUT

OUTPUT:

- m_ready → ACCUMULATE, reset count
- !m_ready → stay

16.1.4 Implementation

```
typedef enum logic {
    ACCUMULATE = 1'b0,
    OUTPUT      = 1'b1
} upsize_state_t;

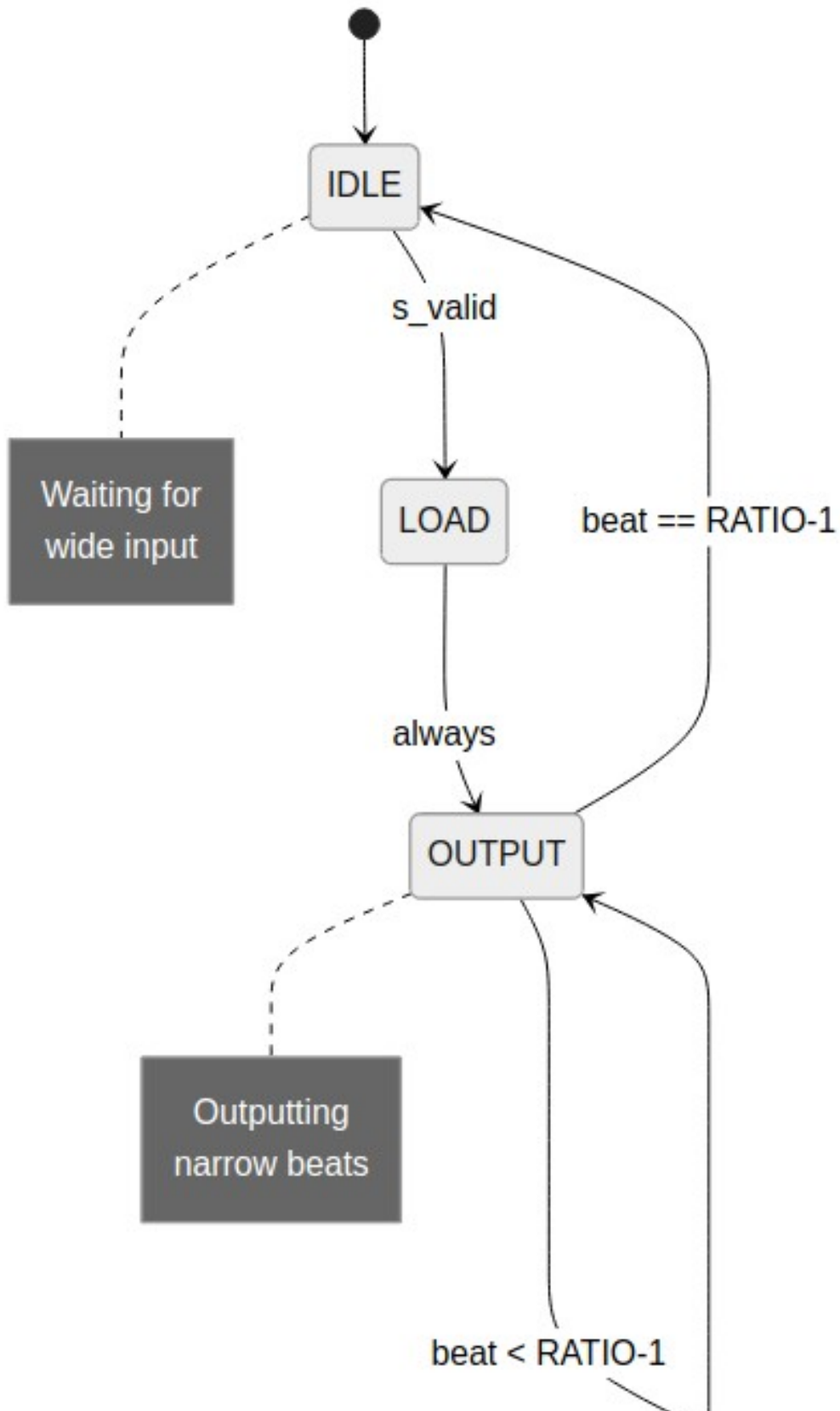
upsize_state_t r_state;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_state <= ACCUMULATE;
    end else begin
        case (r_state)
            ACCUMULATE: begin
                if (s_valid && s_ready) begin
                    if (s_last || r_count == RATIO - 1)
                        r_state <= OUTPUT;
                end
            end
            OUTPUT: begin
                if (m_ready)
                    r_state <= ACCUMULATE;
            end
        endcase
    end
end
```

16.2 4.1.2 Downsize FSM (Single Buffer)

The **axi_data_dnsiz** single-buffer mode uses a load/output state machine.

16.2.1 Figure 4.2: Downsize Single-Buffer FSM



Downsize FSM

16.2.2 States

Table 4.2: Downsize FSM States

State	Description
IDLE	Waiting for wide input
LOAD	Loading wide beat
OUTPUT	Outputting narrow beats

16.2.3 Transitions

IDLE:

- s_valid → LOAD

LOAD:

- always → OUTPUT (combinational)

OUTPUT:

- m_ready && count < RATIO-1 → stay, increment count
- m_ready && count == RATIO-1 → IDLE

16.3 4.1.3 Downsize FSM (Dual Buffer)

Dual-buffer mode uses two parallel state machines with an arbiter.

16.3.1 Buffer State Machine

```
typedef enum logic [1:0] {  
    BUF_EMPTY      = 2'b00,  
    BUF_LOADED     = 2'b01,  
    BUF_OUTPUTTING = 2'b10  
} buf_state_t;
```

```
buf_state_t r_buf_a_state, r_buf_b_state;
```

16.3.2 Arbiter Logic

```
// Select which buffer outputs  
always_comb begin  
    if (r_buf_a_state == BUF_OUTPUTTING)  
        output_sel = 1'b0; // Buffer A  
    else if (r_buf_b_state == BUF_OUTPUTTING)  
        output_sel = 1'b1; // Buffer B  
    else if (r_buf_a_state == BUF_LOADED)  
        output_sel = 1'b0; // A loaded first  
    else  
        output_sel = 1'b1; // B loaded first  
end  
  
// Select which buffer loads  
always_comb begin  
    if (r_buf_a_state == BUF_EMPTY)  
        load_sel = 1'b0;  
    else  
        load_sel = 1'b1;  
end
```

16.4 4.1.4 Full Converter FSMs

16.4.1 Write Converter (axi4_dwidth_converter_wr)

IDLE:

- AW valid → accept AW, store info
- W valid → buffer W data

AW_ACCEPT:

- downstream AW ready → forward adjusted AW

W_CONVERT:

- upsize accumulating narrow W beats
- on output → forward wide W beat

B_FORWARD:

- B from downstream → forward to master

16.4.2 Read Converter (axi4_dwidth_converter_rd)

IDLE:

- AR valid → accept AR, store info, forward adjusted AR

AR_FORWARD:

- downstream AR ready → wait for R

R_CONVERT:

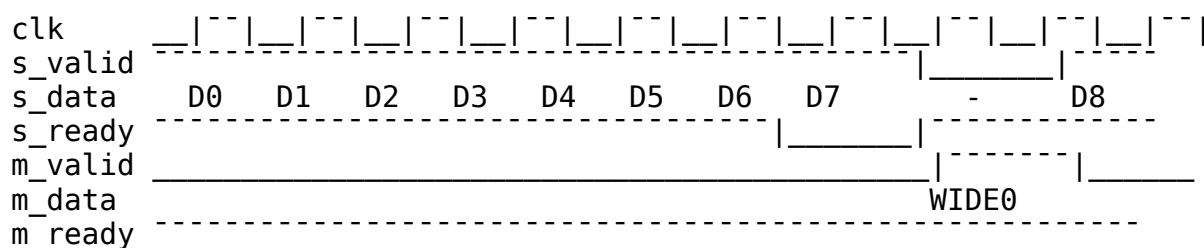
- downsize splitting wide R into narrow beats
- track burst count for RLAST

R_FORWARD:

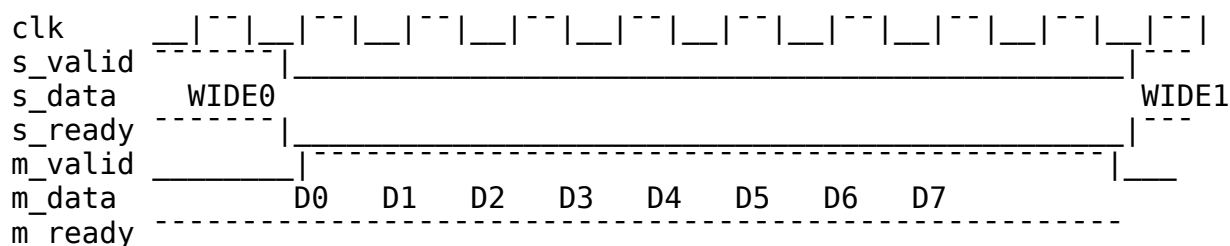
- narrow R beat ready → forward to master
- on RLAST → IDLE

16.5 4.1.5 Timing Diagrams

16.5.1 Upsize Timing (8:1 ratio)



16.5.2 Downsize Timing (8:1 ratio, single buffer)



Next: [Protocol Converter FSMs](#)

17 4.2 Protocol Converter FSMs

These are the state machines inside the protocol converters.

17.1 4.2.1 AXI4 to AXI4-Lite Read FSM

17.1.1 Figure 4.3: AXI4 to AXI4 Read FSM

AXI4 to AXI4 Read FSM

AXI4 to AXI4 Read FSM

17.1.2 States

Table 4.3: AXI4 to AXI4 Read FSM States

State	Description
IDLE	Waiting for AR transaction
SINGLE	Single-beat passthrough
DECOMPOSE	Issuing burst as single beats
WAIT_R	Waiting for final R response

17.1.3 Transitions

IDLE:

- s_arvalid && s_arlen == 0 → SINGLE (passthrough)
- s_arvalid && s_arlen > 0 → DECOMPOSE

SINGLE:

- m_arready → forward AR, wait for R
- m_rvalid && m_rready → IDLE

DECOMPOSE:

- m_arready → issue single AR
- increment address, decrement remaining
- remaining == 0 → WAIT_R

WAIT_R:

- s_rvalid && s_rready && s_rlast → IDLE

17.1.4 Implementation

```
typedef enum logic [1:0] {  
    IDLE      = 2'b00,  
    SINGLE    = 2'b01,  
    DECOMPOSE = 2'b10,  
    WAIT_R    = 2'b11  
} rd_state_t;
```

```
rd_state_t r_state;  
logic [7:0] r_beats_remaining;
```

```

logic [ADDR_WIDTH-1:0] r_current_addr;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_state <= IDLE;
    end else begin
        case (r_state)
            IDLE: begin
                if (s_arvalid && s_arready) begin
                    r_current_addr <= s_araddr;
                    r_beats_remaining <= s_arlen;
                    r_state <= (s_arlen == 0) ? SINGLE : DECOMPOSE;
                end
            end

            SINGLE: begin
                if (m_rvalid && s_rready)
                    r_state <= IDLE;
            end

            DECOMPOSE: begin
                if (m_arvalid && m_arready) begin
                    r_current_addr <= r_current_addr + (1 <<
r_arsize);
                    if (r_beats_remaining == 0)
                        r_state <= WAIT_R;
                    else
                        r_beats_remaining <= r_beats_remaining - 1;
                    end
                end

            WAIT_R: begin
                if (s_rvalid && s_rready && s_rlast)
                    r_state <= IDLE;
            end
        endcase
    end
end

```

17.2 4.2.2 AXI4 to AXI4-Lite Write FSM

17.2.1 States

Table 4.4: AXI4 to AXI4-Lite Write FSM States

State	Description
IDLE	Waiting for AW/W transactions
SYNC_AW_W	Synchronizing AW and W channels
DECOMPOSE	Issuing burst as single beats
WAIT_B	Waiting for all B responses
SEND_B	Sending aggregated B response

17.2.2 AW/W Synchronization

// Track which channels have been accepted

```
logic r_aw_pending, r_w_pending;
```

```
always_ff @(posedge clk) begin
```

```
    // Accept AW
```

```
    if (s_awvalid && s_awready && !r_aw_pending)
        r_aw_pending <= 1'b1;
```

```
    // Accept W
```

```
    if (s_wvalid && s_wready && !r_w_pending)
        r_w_pending <= 1'b1;
```

```
    // Issue AXI4 when both ready
```

```
    if (w_issue_axil4) begin
        r_aw_pending <= 1'b0;
        r_w_pending <= 1'b0;
```

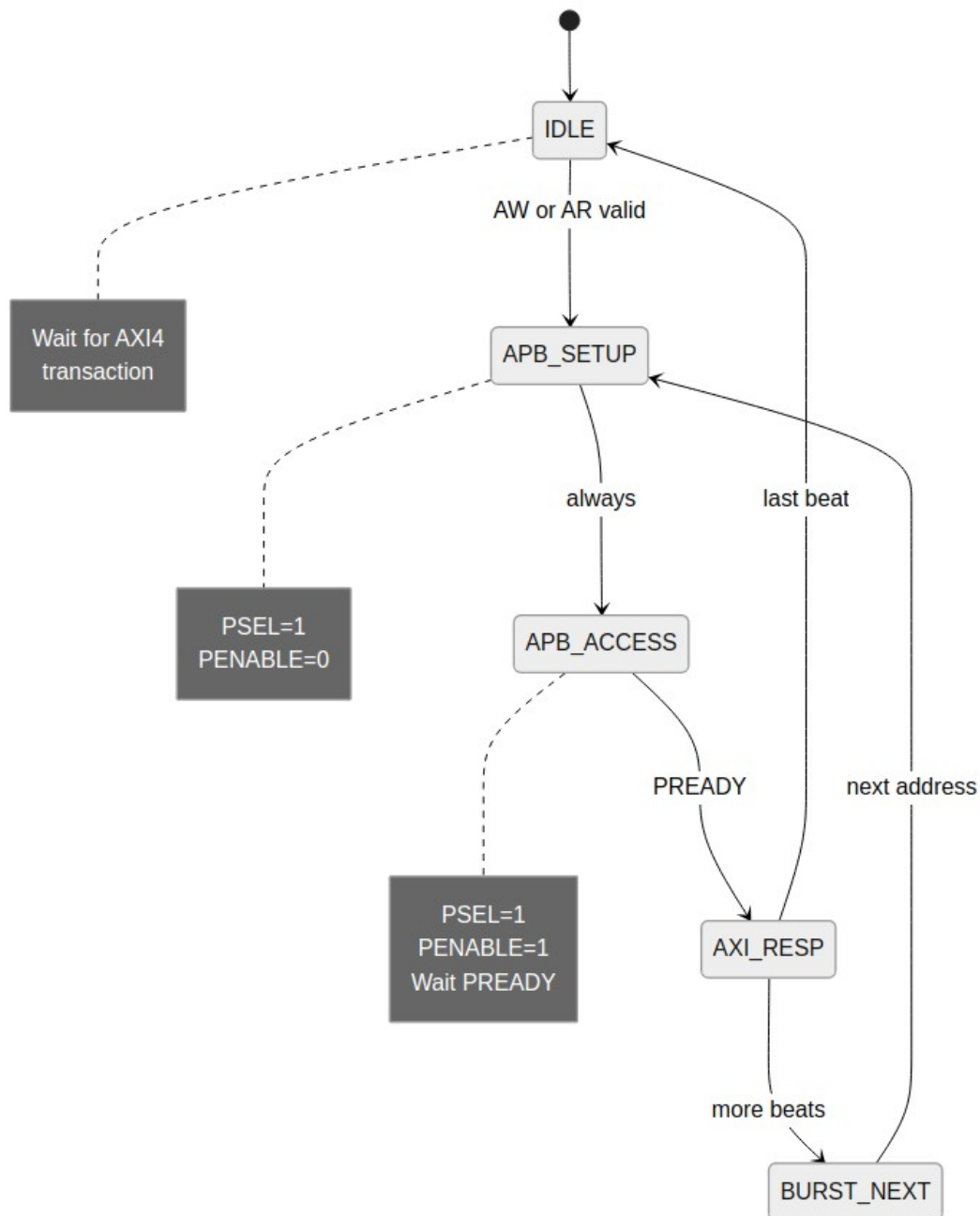
```
    end
```

```
end
```

```
assign w_issue_axil4 = (r_aw_pending || s_awvalid) &&
    (r_w_pending || s_wvalid) &&
    m_awready && m_wready;
```

17.3 4.2.3 AXI4 to APB FSM

17.3.1 Figure 4.4: AXI4 to APB FSM



AXI4 to APB FSM

17.3.2 States

Table 4.5: AXI4 to APB FSM States

State	Description
IDLE	Waiting for AXI4 transaction
APB_SETUP	APB setup phase (PSEL=1, PENABLE=0)
APB_ACCESS	APB access phase (PSEL=1, PENABLE=1)
AXI_RESP	Sending AXI4 response
BURST_NEXT	Preparing next beat in burst

17.3.3 APB Protocol Phases

Setup Phase:

- PSEL = 1
- PENABLE = 0
- PADDR, PWDATA, PWRITE stable
- Duration: 1 cycle

Access Phase:

- PSEL = 1
- PENABLE = 1
- Wait for PREADY
- Sample PRDATA (read) or complete write
- Duration: 1+ cycles (depends on PREADY)

17.3.4 Implementation

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_state <= IDLE;
        psel <= 1'b0;
        penable <= 1'b0;
    end else begin
        case (r_state)
            IDLE: begin
                psel <= 1'b0;
                penable <= 1'b0;
                if (s_awvalid || s_arvalid) begin
                    r_state <= APB_SETUP;
                    psel <= 1'b1;
                end
            end
        end
    end
end
```

```

APB_SETUP: begin
    r_state <= APB_ACCESS;
    penable <= 1'b1;
end

APB_ACCESS: begin
    if (pready) begin
        psel <= 1'b0;
        penable <= 1'b0;
        r_state <= AXI_RESP;
    end
end

AXI_RESP: begin
    if ((r_is_write && s_bready) ||
        (!r_is_write && s_rready)) begin
        if (r_beat_count == r_burst_len)
            r_state <= IDLE;
        else
            r_state <= BURST_NEXT;
        end
    end
end

BURST_NEXT: begin
    r_beat_count <= r_beat_count + 1;
    r_current_addr <= r_current_addr + (1 << r_size);
    r_state <= APB_SETUP;
    psel <= 1'b1;
end
endcase
end
end
end

```

17.4 4.2.4 Timing Analysis

17.4.1 AXI4 to AXIL4 Timing

Single-beat transaction:

Cycle 0: AR accepted
Cycle 1: AR forwarded to AXIL4
Cycle 2: R received from AXIL4
Cycle 3: R forwarded to AXI4
Total: ~2 cycles overhead (can be pipelined)

N-beat burst:

Cycle 0: AR accepted
Cycles 1-2N: Decomposed transactions
Total: 2N cycles

17.4.2 AXI4 to APB Timing

Single transfer:

Cycle 0: AXI4 AR/AW accepted
Cycle 1: APB setup phase
Cycle 2+: APB access phase (wait PREADY)
Cycle N: APB complete, AXI4 response
Total: 3+ cycles (minimum)

Next: [Burst Decomposition](#)

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18 4.3 Burst Decomposition

Burst handling is where the two converter families diverge: width converters rescale the burst length, protocol converters split the burst into single beats.

18.1 4.3.1 Width Converter Burst Handling

18.1.1 Burst Length Adjustment

When converting widths, burst length changes inversely with data width:

$$M_AWLEN = (S_AWLEN + 1) / \text{RATIO} - 1$$

Example (64-bit to 512-bit, $\text{RATIO}=8$):

$$S_AWLEN = 7 \text{ (8 beats} \times 64 \text{ bits} = 512 \text{ bits)}$$

$$M_AWLEN = (7 + 1) / 8 - 1 = 0 \text{ (1 beat} \times 512 \text{ bits)}$$

$$S_AWLEN = 15 \text{ (16 beats} \times 64 \text{ bits} = 1024 \text{ bits)}$$

$$M_AWLEN = (15 + 1) / 8 - 1 = 1 \text{ (2 beats} \times 512 \text{ bits)}$$

18.1.2 Figure 4.5: Width Burst Conversion

Width Burst Conversion

Width Burst Conversion

18.1.3 Non-Aligned Bursts

When the burst length is not a multiple of the ratio:

$$S_AWLEN = 5 \text{ (6 beats)}, \text{RATIO} = 8$$

$$M_AWLEN = (5 + 1) / 8 - 1 = -1 \rightarrow 0 \text{ (1 beat)}$$

The 6 narrow beats pack into 1 wide beat.
Last 2 positions have $\text{WSTRB} = 0$ (no write).

18.2 4.3.2 Protocol Converter Burst Handling

18.2.1 AXI4 to AXI4-Lite Decomposition

AXI4-Lite only supports single-beat transactions, so all bursts must be decomposed:

AXI4 Burst:

ARADDR = 0x1000
ARLEN = 3 (4 beats)
ARSIZE = 2 (4 bytes)

AXI4 Sequence:

Transaction 0: ARADDR = 0x1000
Transaction 1: ARADDR = 0x1004
Transaction 2: ARADDR = 0x1008
Transaction 3: ARADDR = 0x100C

18.2.2 Address Increment Calculation

```
// Calculate address increment based on burst type and size
function automatic [ADDR_WIDTH-1:0] next_address(
    input [ADDR_WIDTH-1:0] current_addr,
    input [2:0] size,
    input [1:0] burst,
    input [7:0] len,
    input [7:0] beat
);
    logic [ADDR_WIDTH-1:0] increment;
    logic [ADDR_WIDTH-1:0] wrap_mask;

    increment = 1 << size;

    case (burst)
        2'b00: // FIXED
            return current_addr; // No increment

        2'b01: // INCR
            return current_addr + increment;

        2'b10: // WRAP
            wrap_mask = ((len + 1) << size) - 1;
            return (current_addr & ~wrap_mask) |
                ((current_addr + increment) & wrap_mask);

    default:
        return current_addr + increment;
```

```
endcase  
endfunction
```

18.3 4.3.3 Response Aggregation

18.3.1 Read Response Aggregation

For burst reads decomposed into multiple single reads:

```
// Track responses as they arrive
logic [7:0] r_response_count;
logic [1:0] r_worst_rresp;

always_ff @(posedge clk) begin
    if (start_new_burst) begin
        r_response_count <= '0;
        r_worst_rresp <= 2'b00; // OKAY
    end else if (m_rvalid && m_rready) begin
        r_response_count <= r_response_count + 1;
        // Keep worst response
        if (m_rresp > r_worst_rresp)
            r_worst_rresp <= m_rresp;
    end
end

// Generate RLAST on final beat
assign s_rlast = (r_response_count == r_original_arlen);

// Forward individual responses or aggregate
assign s_rresp = m_rresp; // Forward each response
// Or: assign s_rresp = r_worst_rresp; // Aggregate
```

18.3.2 Write Response Aggregation

For burst writes:

```
// Track worst response across burst
logic [1:0] r_worst_bresp;
logic r_all_beats_done;

always_ff @(posedge clk) begin
    if (start_new_burst)
        r_worst_bresp <= 2'b00;
    else if (m_bvalid && m_bready)
        r_worst_bresp <= (m_bresp > r_worst_bresp) ? m_bresp :
r_worst_bresp;
end

// Send single aggregated B response
```

```
assign s_bvalid = r_all_beats_done;  
assign s_bresp = r_worst_bresp;
```

18.4 4.3.4 Burst Tracking Registers

18.4.1 Required State

```
// Burst tracking registers
logic [ADDR_WIDTH-1:0] r_base_addr;
logic [ADDR_WIDTH-1:0] r_current_addr;
logic [7:0] r_original_len;
logic [7:0] r_remaining_beats;
logic [2:0] r_size;
logic [1:0] r_burst;
logic [ID_WIDTH-1:0] r_id;
logic r_is_write;
```

18.4.2 Initialization

```
always_ff @(posedge clk) begin
    if (accept_new_transaction) begin
        r_base_addr <= s_axaddr;
        r_current_addr <= s_axaddr;
        r_original_len <= s_axlen;
        r_remaining_beats <= s_axlen;
        r_size <= s_axsize;
        r_burst <= s_axburst;
        r_id <= s_axid;
        r_is_write <= is_write_transaction;
    end else if (beat_complete) begin
        r_current_addr <= next_address(...);
        r_remaining_beats <= r_remaining_beats - 1;
    end
end
```

18.5 4.3.5 Timing Impact

18.5.1 Decomposition Overhead

Table 4.6: Decomposition Overhead

Transaction Type	Overhead
Single-beat	0 cycles (passthrough)
2-beat burst	2 cycles (sequential)
N-beat burst	2N cycles (2 per beat)

18.5.2 Pipeline Considerations

Decomposition is inherently sequential: - Cannot issue next AR until previous R received (AXIL4) - Cannot issue next AW/W until previous B received (AXIL4)

Optimization: Use response pipelining when downstream supports it.

Next: [Chapter 5: Verification](#)

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19 5.1 Test Strategy

The verification approach for the converter modules, from unit tests to system stress.

19.1 5.1.1 Test Organization

19.1.1 Test Hierarchy

```
projects/components/converters/dv/tests/  
├── width_converters/  
│   ├── test_axi_data_upsize.py  
│   ├── test_axi_data_dsize.py  
│   ├── test_axi4_dwidth_converter_wr.py  
│   └── test_axi4_dwidth_converter_rd.py  
├── protocol_converters/  
│   ├── test_axi4_to_axil4_rd.py  
│   ├── test_axi4_to_axil4_wr.py  
│   ├── test_axil4_to_axi4_rd.py  
│   ├── test_axil4_to_axi4_wr.py  
│   ├── test_axi4_to_apb4.py  
│   └── test_peakrdl_adapter.py  
└── integration/  
    ├── test_width_protocol_chain.py  
    └── test_full_system.py
```

19.2 5.1.2 Test Levels

19.2.1 Level 1: Unit Tests

Purpose: Verify individual module functionality in isolation.

Coverage: - All parameters combinations - Edge cases (min/max values) - Error injection

Example:

```
@pytest.mark.parametrize("narrow_width,wide_width", [
    (32, 64),
    (32, 128),
    (64, 256),
    (64, 512),
    (128, 1024),
])
async def test_upsize_ratios(dut, narrow_width, wide_width):
    """Test various width ratios."""
    tb = UpsizeTB(dut, narrow_width, wide_width)
    await tb.reset()
    await tb.run_basic_transfer(count=100)
    assert tb.scoreboard.check_passed()
```

19.2.2 Level 2: Integration Tests

Purpose: Verify module combinations work together.

Coverage: - Width converter + protocol converter chains - Back-to-back converters - Mixed traffic patterns

19.2.3 Level 3: System Tests

Purpose: Verify in realistic system context.

Coverage: - CPU-to-DDR paths - Peripheral access paths - Full bandwidth stress

19.3 5.1.3 Test Categories

19.3.1 Functional Tests

Table 5.1: Functional Test Categories

Category	Description	Example
Basic	Single transactions	Single read, single write
Burst	Multi-beat transactions	INCR burst, WRAP burst
Mixed	Interleaved R/W	Read-modify-write sequences
Edge	Boundary conditions	Max burst length, min width

19.3.2 Stress Tests

Table 5.2: Stress Test Categories

Category	Description	Duration
Throughput	Maximum bandwidth	10,000 transactions
Backpressure	Ready signal variations	Random delays
Reset	Reset during operation	Mid-transaction reset

19.3.3 Error Tests

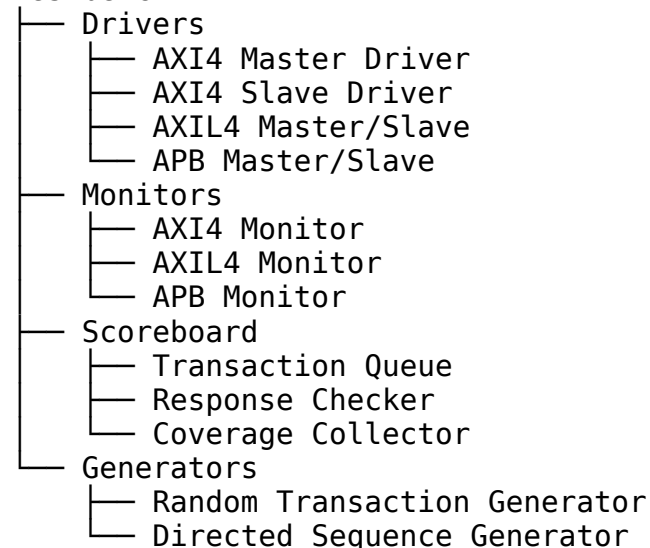
Table 5.3: Error Test Categories

Category	Description	Expected Behavior
SLVERR	Slave error injection	Error propagation
DECERR	Decode error	Error response
Timeout	Response timeout	Error handling

19.4 5.1.4 Testbench Architecture

19.4.1 Components

Testbench



19.4.2 Driver Implementation

```
class AXI4MasterDriver:
    def __init__(self, dut, clock, prefix="s_axi"):
        self.dut = dut
        self.clock = clock
        self.prefix = prefix

    async def write(self, addr, data, burst_len=0):
        """Issue AXI4 write transaction."""
        # AW phase
        self.dut.awvalid.value = 1
        self.dut.awaddr.value = addr
        self.dut.awlen.value = burst_len
        await self._wait_ready("awready")

        # W phase
        for i, d in enumerate(data):
            self.dut.wvalid.value = 1
            self.dut.wdata.value = d
            self.dut.wlast.value = (i == len(data) - 1)
            await self._wait_ready("wready")

        # B phase
```

```
await self._wait_valid("bvalid")
return self.dut.bresp.value
```

19.5 5.1.5 Coverage Model

19.5.1 Functional Coverage

```
@coverage
class ConverterCoverage:
    # Width ratio coverage
    ratio_cp = coverpoint(
        lambda: self.width_ratio,
        bins=[2, 4, 8, 16]
    )

    # Burst length coverage
    burst_len_cp = coverpoint(
        lambda: self.burst_len,
        bins=list(range(0, 256, 16)) + [255]
    )

    # Burst type coverage
    burst_type_cp = coverpoint(
        lambda: self.burst_type,
        bins=["FIXED", "INCR", "WRAP"]
    )

    # Cross coverage
    ratio_x_burst = cross(ratio_cp, burst_len_cp)
```

19.5.2 Code Coverage

Target: >95% line coverage, >90% branch coverage

Table 5.4: Coverage Targets

Module	Line	Branch	FSM
axi_data_upsiz e	98%	95%	100%
axi_data_dnsiz e	97%	93%	100%
axi4_to_axil4_r d	96%	91%	100%
axi4_to_axil4_ wr	95%	90%	100%

19.6 5.1.6 Test Execution

19.6.1 Running Tests

Run all converter tests

```
cd projects/components/converters/dv/tests
pytest -v
```

Run specific module tests

```
pytest test_axi_data_upsize.py -v
```

Run with coverage

```
pytest --cov=projects/components/converters/rtl -v
```

Run with waveform dump

```
pytest test_axi_data_upsize.py -v --waves
```

19.6.2 CI/CD Integration

```
converter_tests:
  stage: test
  script:
    - cd projects/components/converters/dv/tests
    - pytest -v --junitxml=results.xml
    - pytest --cov=rtl --cov-report=html
  artifacts:
    reports:
      junit: results.xml
  paths:
    - htmlcov/
```

Next: [Debug Guide](#)

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20 5.2 Debug Guide

Start here when a converter misbehaves.

20.1 5.2.1 Common Issues

20.1.1 Width Converter Issues

20.1.1.1 *Issue: Data Corruption*

Symptoms: - Output data doesn't match expected - Random bits flipped or missing

Debug Steps: 1. Check width ratio calculation: $RATIO = WIDE_WIDTH / NARROW_WIDTH$ 2. Verify beat counter is correct: $\$clog2(RATIO)$ bits 3. Check data packing/unpacking slice indices 4. Verify sideband mode matches use case

Waveform Checkpoints:

r_count	- Should cycle 0 to RATIO-1
r_data	- Check each slice is populated
s_data/m_data	- Compare input/output patterns

20.1.1.2 *Issue: LAST Signal Incorrect*

Symptoms: - Transaction ends early or late - Master receives wrong beat count

Debug Steps: 1. Check USE_LAST parameter 2. Verify burst tracker logic if enabled 3. Check s_last input timing

Solution:

```
// Verify LAST generation
assign m_last = (r_count == RATIO - 1) || r_input_last_seen;
```

20.1.1.3 *Issue: Throughput Lower Than Expected*

Symptoms: - Gaps between output beats - Single-buffer achieving <80%

Debug Steps: 1. Check downstream ready signal behavior 2. Verify DUAL_BUFFER parameter for downsize 3. Look for backpressure stalls

20.1.2 Protocol Converter Issues

20.1.2.1 *Issue: Burst Decomposition Incorrect*

Symptoms: - Wrong number of single transactions - Address increment wrong

Debug Steps: 1. Check burst type (FIXED, INCR, WRAP) 2. Verify size calculation 3. Check address increment logic

Waveform Checkpoints:

r_current_addr	- Should increment by $(1 \ll \text{size})$
r_beat_count	- Should count to s_arlen/s_awlen
m_arvalid	- Should assert for each decomposed beat

20.1.2.2 *Issue: Response Aggregation Wrong*

Symptoms: - Wrong BRESP/RRESP value - Error not propagated correctly

Debug Steps: 1. Check worst-case response tracking 2. Verify response counter 3. Check RLAST generation

Solution:

```
// Proper error aggregation
always_ff @(posedge clk) begin
    if (new_burst)
        r_worst_resp <= 2'b00;
    else if (m_rvalid && m_rready)
        r_worst_resp <= (m_rresp > r_worst_resp) ? m_rresp :
r_worst_resp;
end
```

20.2 5.2.2 Debug Signals

20.2.1 Recommended Internal Signals

For width converters:

```
// Add debug outputs
output logic [$clog2(RATIO)-1:0] dbg_beat_count,
output logic                      dbg_buffer_valid,
output logic [1:0]                dbg_state
```

For protocol converters:

```
// Add debug outputs
output logic [7:0]  dbg_remaining_beats,
output logic [2:0]  dbg_state,
output logic [1:0]  dbg_worst_resp,
output logic        dbg_in_burst
```

20.2.2 ILA Configuration

```
# Create ILA for converter debug
create_debug_core u_ila ila
```

```
# Add probes
set_property probe_count 10 [get_debug_cores u_ila]
connect_debug_port u_ila/clk [get_nets aclk]
```

```
# Key signals
connect_debug_port u_ila/probe0 [get_nets r_state]
connect_debug_port u_ila/probe1 [get_nets r_beat_count]
connect_debug_port u_ila/probe2 [get_nets s_arvalid]
connect_debug_port u_ila/probe3 [get_nets m_arvalid]
```

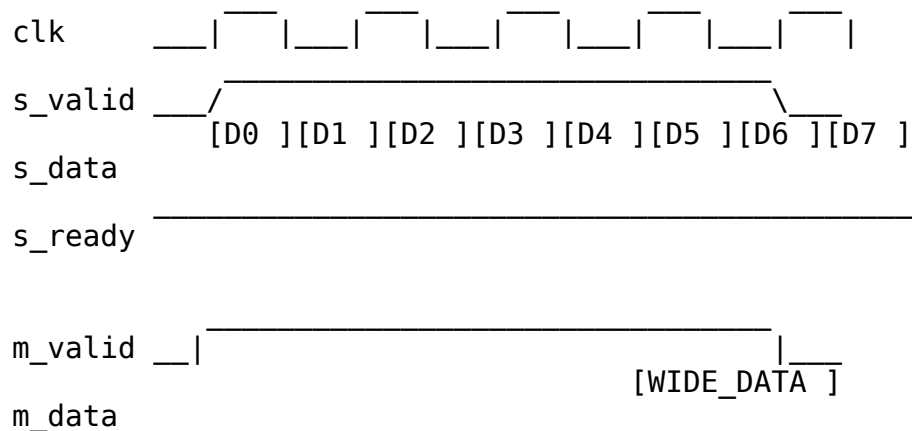
20.3 5.2.3 Simulation Debug

20.3.1 Waveform Analysis

Key Signal Groups:

1. **Input Channel:**
 - s_valid, s_ready, s_data, s_last
2. **Output Channel:**
 - m_valid, m_ready, m_data, m_last
3. **Control:**
 - r_state, r_count, r_burst_remaining
4. **Sideband:**
 - s_wstrb/m_wstrb (write path)
 - s_rresp/m_rresp (read path)

20.3.2 Timing Diagram Template



Check:

1. s_ready stays high during accumulation
2. m_valid asserts after RATIO beats
3. Data packing is correct

20.4 5.2.4 Common Mistakes

20.4.1 Mistake 1: Wrong Width Ratio

```
// WRONG: Manual ratio  
localparam RATIO = 8; // May not match actual widths
```

```
// CORRECT: Calculated ratio  
localparam RATIO = WIDE_WIDTH / NARROW_WIDTH;
```

20.4.2 Mistake 2: Missing Sideband Handling

```
// WRONG: Forgetting sideband  
assign m_data = r_data;  
// Missing: assign m_wstrb = ...
```

```
// CORRECT: Handle both  
assign m_data = r_data;  
assign m_wstrb = r_sideband;
```

20.4.3 Mistake 3: Incorrect LAST Timing

```
// WRONG: LAST on wrong beat  
assign m_last = r_count == 0; // First beat!  
  
// CORRECT: LAST on final beat  
assign m_last = (r_count == RATIO - 1) || r_early_last;
```

20.4.4 Mistake 4: Burst Length Calculation Error

```
// WRONG: Off-by-one  
assign m_awlen = s_awlen / RATIO; // Wrong!  
  
// CORRECT: Account for LEN encoding  
assign m_awlen = ((s_awlen + 1) >> RATIO_LOG2) - 1;
```

20.5 5.2.5 Verification Checklist

Before signoff, verify:

- ☐ All parameter combinations tested
- ☐ Single-beat transactions work
- ☐ Multi-beat bursts work (INCR, WRAP, FIXED)
- ☐ Backpressure handling correct
- ☐ Error propagation correct
- ☐ LAST signal timing correct
- ☐ Sideband signals handled correctly
- ☐ Reset behavior verified
- ☐ Coverage targets met

20.6 5.2.6 Performance Validation

20.6.1 Throughput Measurement

```
async def measure_throughput(tb, transaction_count=1000):
    start_time = get_sim_time()

    for _ in range(transaction_count):
        await tb.send_transaction()

    end_time = get_sim_time()
    elapsed_cycles = (end_time - start_time) / clock_period

    throughput = transaction_count / elapsed_cycles
    print(f"Throughput: {throughput:.2f} transactions/cycle")

    return throughput
```

20.6.2 Expected Throughput

Table 5.5: Expected Throughput

Module	Mode	Expected
axi_data_upsize	Single	1.0 trans/cycle
axi_data_dnsize	Single	0.8 trans/cycle
axi_data_dnsize	Dual	1.0 trans/cycle
axi4_to_axil4	Single-beat	1.0 trans/cycle
axi4_to_axil4	Burst	0.5 trans/cycle

End of Micro-Architecture Specification